Limits

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Limits

INTRODUCTION TO LIMITS

DEFINITION OF LIMITS :

We can also say that if f(x) becomes arbitrarily close to a unique number L as x approaches 'c' from either side, then the **limit** of f(x) as x approaches 'c' is L. This is written as:

$$\lim_{x \to c} f(x) = L$$

Let us take an example:

$$y = \frac{x^2 - 4}{x - 2} = x + 2; x \neq 2$$

Here, f(2) is not defined as it is of the $\frac{0}{0}$ form which is an indeterminate form.

But as $x \to 2$, $f \to 4$

which means if x is sufficiently close to 2, then f is arbitrarily close to 4.

Here, solving limit leads to dividing by (x – 2) (Really!)
In above example, $x \to 2$ means x is very close to 2 like 2.000000 ... 00001 or 1.9999 ... 999999.

Similarly, $x \to 2 \Rightarrow f \to 4$ which means when x becomes very close to 2, then f becomes very close to 4 like 4.00000000 ... 0001 or 3.99999 ... 99.

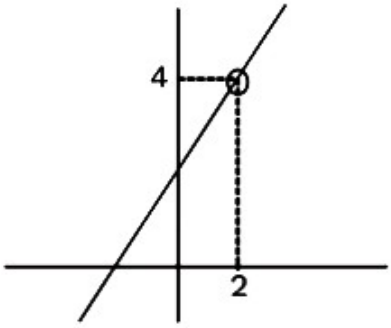
In such case it is said that limit of function at x = 2 is 4. It is represented by $\lim_{x \to 2} f(x) = 4$ (approx. value).

Now **SOLVING TRICK** for $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$:

- Since it is of the form $\frac{0}{0}$ at x=2, cancel out the common factor x-2 So, that it is no longer of the form $\frac{0}{0}$.

Definition

Limit of a function at any point can be said to be the value of the function at the immediate neighbourhood of that point.



$$\lim_{x\to 2}\frac{(x-2)(x+2)}{x-2}=\lim_{x\to 2}(x+2)=4$$

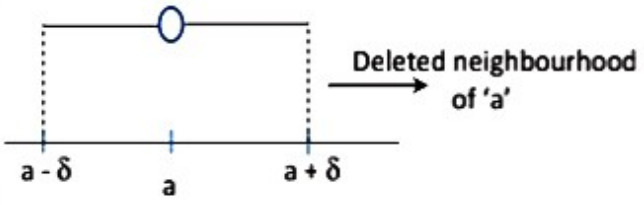
Now, there is no problem in putting the value of x = 2 in the limit.

Deleted neighbourhood of a point 'a'

$(a-\delta, a+\delta)-\{a\}$ is called deleted neighbourhood of 'a' where δ is very small positive.

$(a-\delta, a)$ is left neighbourhood.

$(a, a+\delta)$ is right neighbourhood.



Meaning of small positive is something like 0.00000000000000 ... ∞

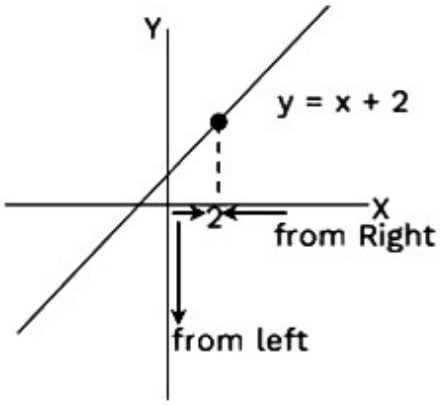
To find limit, function must be defined in neighbourhood of the point.

Only comment on Limit when function is continuous in neighbourhood.

Left and Right Hand limits :

$\lim_{x\to a^-} f(x)=\ell=f(a^-)$ means as x tends to a from left hand side, f tends to ℓ, $x\neq a$.

$\lim_{x\to a^+} f(x)=M=f(a^+)$ means as x tends to a from right hand side, f tends to M, $x\neq a$.



Left Hand limit (L.H.L) here means value of function at left neighbourhood of point.

$$\text{L.H.L}=\lim_{x\to 2^-}(x+2)=4$$

$x\to 2^-$ means x = 2 - 0.000000 ... 001

Similarly, Right Hand limit means value of function at right neighbourhood of point.

R.H.L = $\lim_{x \to 2^+} (x+2) = 4$

$x \to 2^+$ means x=2 + 0.000000 ... 0001.

Existence of limit :

If L.H.L = R.H.L = finite value M, then limit exists and is equal to 'M'.

Or $f(a^{-1}) = f(a^+) =$ finite = Limit

In previous example, L.H.L = 4 = R.H.L at x = 2.

So, limit exists at x = 2 and $\lim_{x \to 2}(x+2) = 4$

7 INDETERMINATE FORMS

$\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty, \infty - \infty, 1^\infty, 0^0, \infty^0$

If $\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0$, then $\lim_{x \to a} \frac{f}{g}$ take the form of $\frac{0}{0}$.

Sometimes this limit exists and has finite value. The determination of limits in such case is referred to as evaluation of indeterminate form.

Some types of **Determinate form** $\Rightarrow \infty + \infty = \infty$; $0^\infty = 0$

Some examples of **Indeterminate form**:

$\frac{x^2-4}{x-2}\left(\frac{0}{0} \text{ form}\right)$, $\frac{x^2}{x-2} - \frac{4}{x-2}(\infty - \infty \text{ form})$,

$(x^2-4)\left(\frac{1}{x-2}\right)(0 \times \infty \text{ form})$, $\frac{\frac{1}{x-2}}{\frac{1}{x^2-4}}\left(\frac{\infty}{\infty} \text{ form}\right)$

All the above examples are for **x = 2**.

NOTE :

(i) $\lim_{x \to a} f(x) \Rightarrow x \neq a$

(ii) $\lim_{x \to a} f(x)$ is said to exits at x = a if

$$\lim_{h\to 0} f(a+h) = \lim_{h\to 0} f(a-h)$$

= (a finite quantity, disregards whether f is defined at x = a)

$f(a^+)$ (R.H.L) $\qquad$ $f(a^-)$ (L.H.L)

TYPES OF INDETERMINATE FORMS

1. $\frac{0}{0}$	**2.** $\frac{\infty}{\infty}$	**3.** $0 \times \infty$
4. $\infty - \infty$	**5.** 0^0	**6.** 1^∞
7. ∞^0	Indeterminate form→ that value of the limit is not obvious	

Ex. $\lim_{x\to -3} \frac{x^2+x-6}{x+3} = \frac{0}{0}$ $\left(\frac{9-3-6}{-3+3} = \frac{0}{0}\right)$

→ Cannot be determined.

The above result has no meaning as a real number. It is called an indeterminate form because you cannot, from the form alone, determine the limit.

Make a table to estimate the above limit.

What is your estimate?

The limit is −5.

The solution is as follows:

$$\lim_{x\to -3} \frac{(x+3)(x-2)}{x+3} = \lim_{x\to -3} (x-2) = -3-2 = -5$$

Checking if limit Exists

To check if limit exists for f(x) at x = a

We check if :

Left hand limit = Right hand Limit = L

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L$$

Recall Notation

$$\lim_{x \to a} f(x) = L$$

This means:

As x gets closer to a, f(x) gets closer to L

Determinate forms :

Let $a \in R$

(1) $a + \infty = + \infty$

(2) $a - \infty = -\infty$

(3) $+\infty + \infty = + \infty$

(4) $-\infty - \infty = -\infty$

(5) $a \times (+\infty) = \begin{cases} +\infty & \text{if} \quad a > 0 \\ -\infty & \text{if} \quad a < 0 \end{cases}$ where $a \neq 0$

(6) $a \times (-\infty) = \begin{cases} +\infty & \text{if} \quad a < 0 \\ -\infty & \text{if} \quad a > 0 \end{cases}$ where $a \neq 0$

(7) $\frac{a}{+\infty} = 0$ and $\frac{a}{-\infty} = 0$

(8) $\frac{a}{0} = \begin{cases} +\infty & \text{if} \quad a > 0 \\ -\infty & \text{if} \quad a < 0 \end{cases}$ where $a \neq 0$

Example :

$$\lim_{x \to 0} \frac{2}{x} = \ ?$$

$$\lim_{x \to 0^+} \frac{2}{x} = +\infty \qquad \left(\frac{2}{0.00000\ldots\ 0001} \right)$$

$$\lim_{x \to 0^-} \frac{2}{x} = -\infty \qquad \left(\frac{2}{-0.0000\ldots\ 001} \right)$$

$\therefore$ L.H.L $\neq$ R.H.L. So limit does not exit.

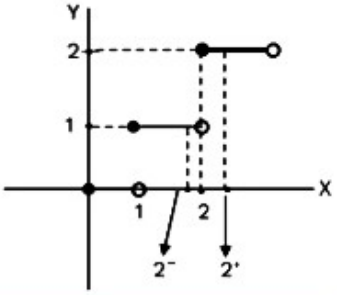
Q.1 **f(x) = [x], find $\lim_{x\to 2}[x]$.**

Sol. (1) LHL = $\lim_{x\to 2^-}[x] = 1$

(2) RHL = $\lim_{x\to 2^+}[x] = 2$

$\Rightarrow$ LHL $\neq$ RHL

$\therefore$ Limit does not exist.



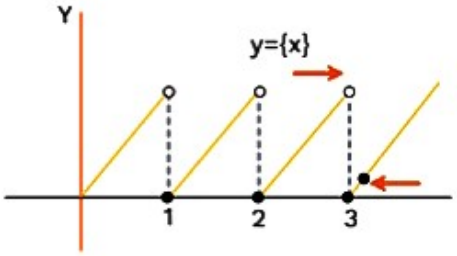
Q.2 $\lim_{x\to 3}\{x\} = ?$

Sol. LHL = $\lim_{x\to 3^-}\{x\} = 1$

RHL = $\lim_{x\to 3^+}\{x\} = 0$

$\therefore$ LHL $\neq$ RHL

So, limit does not exist at x = 3.



Note :

(1) $\lim_{x\to}[x]$ does not exist.

($\because$ RHL = I, LHL = I - 1)

(2) $\lim_{x\to I}\{x\}$ does not exist.

($\because$ RHL = 0, LHL = 1)

Q.3 $f(x) = \frac{|x|}{x}$. **Find** $\lim_{x\to 0} f(x)$.

Sol. LHL = $\lim_{x\to 0^-}\frac{|x|}{x} = \lim_{x\to 0^-}\frac{-x}{x} = -1$

RHL = $\lim_{x\to 0^+}\frac{|x|}{x} = \lim_{x\to 0^-}\frac{x}{x} = 1$

$\Rightarrow$ LHL $\neq$ RHL

So, limit does not exist.

Q.4 $\lim\limits_{x\to 0} \tan^{-1}\frac{1}{x}$

Sol. $\lim\limits_{x\to 0^+} \tan^{-1}\left(\frac{1}{x}\right) = \tan^{-1}(\infty) = \frac{\pi}{2} = \text{RHL}$

$\lim\limits_{x\to 0^-} \tan^{-1}\left(\frac{1}{x}\right) = \tan^{-1}(-\infty) = \frac{-\pi}{2} = \text{LHL}$

$\therefore$ LHL $\neq$ RHL. Limit does not exist.

Q.5 $\lim\limits_{x\to 0} \sin^{-1}(\sec x)$

Sol. LHL $= \lim\limits_{x\to 0^-} \sin^{-1}(\sec x)$

But $\sec(0^-) > 1$ and domain of $\sin^{-1}x$ is $[-1, 1]$.

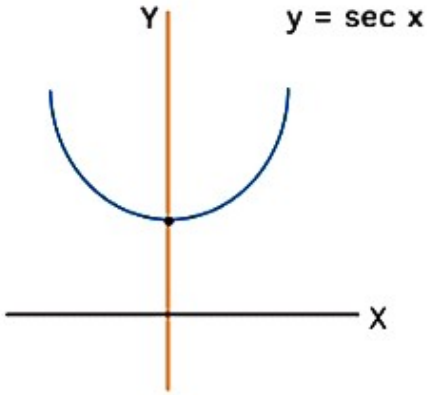
$\therefore$ $\sin^{-1}(\sec 0^-)$ will not exist.

Similarly, for RHL, $\sin^{-1}(\sec 0^+)$ will not exist.

$\therefore$ Both LHL and RHL does not exist.

So, limit does not exist.

as $x\to 0$, $\sec\to 1^+$ and $\sin^{-1}(1^+)$ does not exist.



Ex: Same as above question, $\lim\limits_{x\to 0} \sin^{-1}(x^2 + 1)$

does not exist. As $x \to 0$, $x^2 + 1 > 1$

So, $\sin^{-1}(x^2 + 1)$ will not exist.

Q.6 $\lim\limits_{x\to 0} \cot^{-1}x^2$

Sol. $\text{LHL} = \lim\limits_{x\to 0^-} \cot^{-1}(x^2) = \frac{\pi}{2}$

$\text{RHL} = \lim\limits_{x\to 0^+} \cot^{-1}(x^2) = \frac{\pi}{2}$

$\therefore$ $\text{LHL} = \text{RHL} = \frac{\pi}{2} = \text{limit}$

Q.7 $\lim_{x\to 1} f(x) = [x] + \sqrt{\{x\}}$

Sol. R.H.L. $= 1 + \sqrt{0} = 1$

L.H.L. $= 0 + \sqrt{1} = 1$

LHL = RHL = 1

So $\lim_{x\to 1} f(x) = 1$

Q.8 $\lim_{x\to 1} x\,\text{sgn}(x-1)$

Sol. As $x \to 1^+ \to x - 1 > 0 \quad \Rightarrow \quad \text{sgn}(x-1) = 1$

$\therefore$ RHL $= 1 \times 1 = 1$

As $x \to 1^- \to x - 1 < 0 \quad \Rightarrow \quad \text{sgn}(x-1) = -1$

$\therefore$ LHL $= 1 \times (-1) = -1$

$\Rightarrow$ LHL $\neq$ RHL, so limit does not exist.

Q.9 $\lim_{x\to \frac{\pi}{2}} [\sin x]$

Sol. $x \to \frac{\pi^-}{2} \Rightarrow \sin x = 1^- \Rightarrow [1^-] = 0$

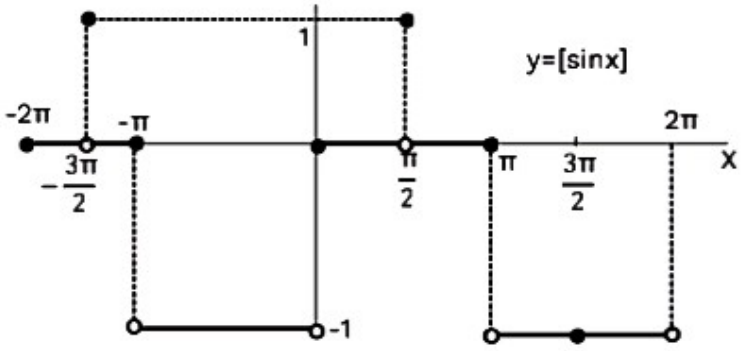
$\Rightarrow$ LHL = 0

$x \to \frac{\pi^+}{2} \Rightarrow \sin x = 1^- \Rightarrow [1^-] = 0$

$\Rightarrow$ RHL = 0

$\therefore$ LHL = RHL = 0 = Limit

$\lim_{x\to \frac{\pi}{2}} [\sin x] = 0$



Note :

Limit is ABSOLUTE in nature, it gives the EXACT value.

Q.10 $\lim\limits_{x\to\frac{\pi}{2}} [\cos x]$

Sol. $x \to \frac{\pi^-}{2} \Rightarrow \cos x = 0^+$

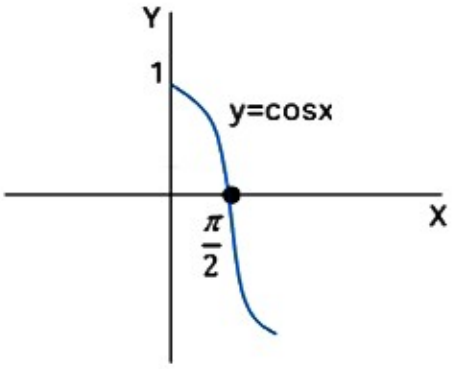
$\Rightarrow [\cos x] = 0 = \text{LHL}$

$x \to \frac{\pi^+}{2} \Rightarrow \cos x = 0^-$

$\Rightarrow [\cos x] = -1 = \text{RHL}$

$\Rightarrow \text{LHL} \neq \text{RHL}$

So, limit does not exist.



One sided limit :

If one side of neighbourhood is not in domain, then the limit of the function will be the limit on other side.

Ex: $\lim\limits_{x\to 0}\sqrt{x}$: → Domain is $[0, \infty)$.

So, left side of function at x=0 does not exist.

So, limit = RHL = $\lim\limits_{x\to 0^+}\sqrt{x} = 0$

Q.1 **Consider** $f(x) = \begin{cases} 2x-3, x \geq 2 \\ 4-x^2, x < 2 \end{cases}$ **. Find** $\lim\limits_{x\to 2} f(x)$.

Sol. $x \to 2^+ \Rightarrow \text{RHL} = 2\times 2 - 3 = 1$

$x \to 2^- \Rightarrow \text{LHL} = 4 - 2^2 = 0$

$\Rightarrow \text{LHL} \neq \text{RHL}$

$\Rightarrow \lim\limits_{x\to 2} f(x)$ does not exist.

Q.2 **Find** $\lim\limits_{x\to 1} \frac{1}{[x]}$

Sol. Domain of $\frac{1}{[x]}$ is $(-\infty, 0) \cup [1, \infty)$

So, limit will be equal to RHL as 1^- is not domain.

$x \to 1^+$

$\Rightarrow [x] = 1$

$\Rightarrow \text{RHL} = \frac{1}{1} = 1$

$\therefore$ Limit = 1

Methods to Evaluate Limit

Various Strategies (To evaluate limit):

Algebraic Methods

Factorisation / Rationalisation / Double rationalisation / Use of binomial theorem / algebraic identities/involving Law of Love if necessary

(We love ∞ in Denominator and 0 in Numerator)

Q.1 $\lim\limits_{n\to\infty} \frac{2n+3}{7n-6} \left(\frac{\infty}{\infty} \text{form}\right)$

Sol. Applying law of love, divide by 'n' in both numerator and denominator.

We get, $\lim\limits_{n\to\infty} \frac{2+\frac{3}{n}}{7-\frac{6}{n}} = \frac{2+0}{7-0} = \frac{2}{7}$ $\left(\because \frac{3}{n} \text{ and } \frac{6}{n} \to 0\right)$

Q.2 $\lim\limits_{x\to 1} \frac{x^3-1}{x-1}$

(A) 0 **(B) 3** **(C) 2** **(D) None of these**

Sol. **(B)**

$\lim\limits_{x\to 1} \frac{x^3-1}{x-1}$ (It is of the $\frac{0}{0}$ form)

$\lim\limits_{x\to 1} \frac{(x-1)(x^2+x+1)}{x-1} = 1+1+1 = 3$

Q.3 $\lim\limits_{x\to 1}\dfrac{x^5-1}{x-1}$

(A) 0 **(B) 3** **(C) 2** **(D) None of these**

Sol. **(D)**

Again, it is of the form $\left(\dfrac{0}{0}\right)$.

So, using factorisation, remove the factor 'x – 1' causing '0'.

$$\Rightarrow \lim_{x\to 1}\frac{(x-1)(1+x+x^2+x^3+x^4)}{x-1} = 1+1+1+1+1 = 5$$

Standard limits :

1. $\lim\limits_{x\to a}\dfrac{x^n-a^n}{x-a} = na^{n-1}; n\in N$

Proof:

$$\frac{x^r-a^n}{x-a} = \frac{a^r\left(\frac{x}{a}\right)^r-1}{a\left(\frac{x}{a}-1\right)} = a^{n-1}\frac{\left(\frac{x}{a}\right)^r-1}{\left(\frac{x}{a}\right)-1}$$

let $\dfrac{x}{a} = t$. As $x\to a \Rightarrow t\to 1$

$$\Rightarrow \lim_{t\to 1} a^{r-1}\left(\frac{t^n-1}{t-1}\right) = na^{n-1}$$ (Same as Previous Question)

Note :

$$\frac{x^r-a^r}{x-a} = x^{n-1}+x^{n-2}a+x^{n-3}a^2+\ldots+xa^{r-2}+a^{n-1}$$

2. $\lim\limits_{x\to a}\dfrac{x^r-a^n}{x-a}$; $n\in Q, n\notin N$

Proof :

Let $x = a + h, h\to 0$

$$\Rightarrow \lim_{h\to 0}\frac{(a+h)^n-a^n}{a+h-a} = a^n\lim_{h\to 0}\frac{\left(1+\frac{h}{a}\right)^n-1}{h}, \frac{h}{a}\to 0$$

$$= a^n \lim_{h\to 0} \frac{\left[1 + n\frac{h}{a} + \frac{n(n-1)}{2}\left(\frac{h}{a}\right)^2 + \ldots\right] - 1}{h}$$

$$= a^n \lim_{h\to 0} \frac{n}{a} + \frac{n(n-1)}{2!}\frac{h}{a} + \frac{n(n-1)(n-2)}{3!}\left(\frac{h}{a}\right)^2 + \ldots$$

$$= na^{n-1}$$

Alternative Method :

Using L' Hospital rule. (used in case of $\frac{0}{0}, \frac{\infty}{\infty}$)

$\lim_{x\to a} \frac{f}{g} = \lim_{x\to a} \frac{f'}{g'} = \lim_{x\to a} \frac{f''}{g''}$ (will be studied later on)

Q.1 $\lim_{x\to 1} \frac{x^{100} - 1}{x^{20} - 1}$

(A) 5 **(B) 3** **(C) 2** **(D) None of these**

Sol. **(A)**

Limit is of $\frac{0}{0}$ form

Using L' Hospital method,

$$\Rightarrow \lim_{x\to 1} \frac{100x^{99}}{20x^{19}} = \frac{100}{20} = 5$$

Alternate method :

$$x^{20} = y \Rightarrow \lim_{y\to 1} \frac{y^5 - 1}{y - 1} = 5$$

Q.2 $\lim_{x\to 1} \frac{x^3 - n}{x - 1}$**. Limit exists. Find n and limit.**

(A) 1, 3 **(B) 3, 1** **(C) 1, 2** **(D) None of these**

Sol. **(A)**

Numerator $\to 1 - n$; Denominator $\to 0$

If $1 - n \neq 0$, limit does not exist.

If $1 - n = 0$, limit may exist if the factors of $x - 1$ get cancel out.

$$\therefore\ n = 1 \Rightarrow \lim_{x\to 1} \frac{x^3 - 1}{x - 1} = 3$$

Q.3 $\lim_{x\to 0}\frac{(\cos x)^{\frac{1}{3}}-(\cos x)^{\frac{1}{2}}}{\sin^2 x}$ **equals**

(A) $\frac{1}{6}$ **(B)** $-\frac{1}{6}$ **(C)** $\frac{1}{12}$ **(D) None of these**

Sol. **(C)**

Let $\cos x = y^6$ so as $x \to 0 \Rightarrow y \to 1$

$$\lim_{y\to 1}\frac{y^2-y^3}{1-y^{12}} = \lim_{y\to 1}\frac{y^2(1-y)}{(1-y^6)(1+y^6)}$$

$$\left(\lim_{y\to 1}\frac{y^2}{1+y^6}\right)\left(\lim_{y\to 1}\frac{(1-y)}{1-y^6}\right) = \frac{1}{2}\times\frac{1}{6} = \frac{1}{12}$$

Q.4 $\lim_{x\to\frac{\pi}{4}}\frac{-1+\cot^3 x}{-2+\cot x+\cot^3 x}$

(A) $\frac{3}{4}$ **(B)** $\frac{4}{3}$ **(C)** $-\frac{3}{4}$ **(D) Does not exist**

Sol. **(A)**

$\cot x = t$ so as $x \to \frac{\pi}{4} \Rightarrow t \to 1$

$$\Rightarrow \lim_{t\to 1}\frac{t^3-1}{t^3+t-2} = \lim_{t\to 1}\frac{t^3-1}{(t-1)(t^2+t+2)}$$

$$\lim_{t\to 1}\left(\frac{t^3-1}{t-1}\right)\left(\frac{1}{t^2+t+2}\right) = 3\times\frac{1}{4} = \frac{3}{4}$$

Q.5 **Find the value of** $\underset{x\to 2}{\text{Limit}}\frac{2^x+2^{3-x}-6}{\sqrt{2^{-x}}-2^{1-x}}$

Sol. $2^x = t$ for $x \to 2 \Rightarrow t \to 4$

$$\Rightarrow \lim_{x\to 4}\frac{t+\frac{8}{t}-6}{\sqrt{\frac{1}{t}}-\frac{2}{t}} = \lim_{t\to 4}\frac{t^2-6t+8}{\sqrt{t}-2}$$

$$= \lim_{x\to 4}\frac{(t-2)(t-4)}{\sqrt{t}-2} = \lim_{t\to 4}\frac{(t-2)(\sqrt{t}+2)(\sqrt{t}-2)}{\sqrt{t}-2} = 2\times 4 = 8$$

Q.6 $\lim\limits_{x\to 0}\dfrac{x}{\sqrt{5-x}-\sqrt{5+x}}$

(A) $\sqrt{5}$ **(B)** $-\sqrt{5}$ **(C)** $2\sqrt{5}$ **(D) None of these**

Sol. **(B)**

Rationalising, we get

$$\lim_{x\to 0}\frac{x\left(\sqrt{5-x}+\sqrt{5+x}\right)}{(5-x)-(5+x)}=\lim_{x\to 0}\frac{x\left(\sqrt{5-x}+\sqrt{5+x}\right)}{-2x}$$

$$=\frac{2\sqrt{5}}{-2}=-\sqrt{5}$$

Q.7 $\lim\limits_{x\to 9}\dfrac{3-\sqrt{x}}{4-\sqrt{2x-2}}$

(A) 1 **(B) 2** **(C) 3** **(D) None of these**

Sol. **(D)**

$$\lim_{x\to 9}\frac{3-\sqrt{x}}{4-\sqrt{2x-2}}\times\left(\frac{3+\sqrt{x}}{3+\sqrt{x}}\right)\times\left(\frac{4+\sqrt{2x-2}}{4+\sqrt{2x-2}}\right)$$ (Double Rationalization)

$$=\lim_{x\to 9}\frac{(9-x)\left(4+\sqrt{2x-2}\right)}{\left(3+\sqrt{x}\right)(18-2x)}$$

$$=\frac{1}{2}\times\frac{(4+4)}{(3+3)}=\frac{2}{3}$$

Q8 $\lim\limits_{x\to 0}x\sin\dfrac{1}{x}$

(A) 1 **(B) 2** **(C) 3** **(D) None of these**

Sol. **(D)**

as $x\to 0$, $\dfrac{1}{x}\to\pm\infty$

In this range, $\sin\dfrac{1}{x}$ will always lie in [-1, 1] which is finite quantity.

Since $x\to 0$, when multiplied by a finite quantity will always give the limit as 0.

So, $\lim\limits_{x\to 0}x\sin\dfrac{1}{x}=0$

Q.9 $\lim_{x\to 1}\cos\left(\frac{\pi}{x+1}\right).\cos\left(\frac{\pi}{x-1}\right)$

(A) 0 **(B) 1** **(C) –1** **(D) None of these**

Sol. **(A)**

As $x\to 1$, $\frac{\pi}{x+1}\to\frac{\pi}{2}\Rightarrow\cos\left(\frac{\pi}{x+1}\right)\to 0$

Again $\cos\left(\frac{\pi}{x-1}\right)\in[-1,1]$

So, $0\times\cos\left(\frac{\pi}{x-1}\right)\to 0 \quad\Rightarrow \lim_{x\to 1}\cos\left(\frac{\pi}{x+1}\right)\cos\left(\frac{\pi}{x-1}\right)=0\times\text{finite}=0$

Q.10 **If** $\lim_{x\to 0}\frac{\sqrt{ax+b}-2}{x}=1$ **, then (a, b) is equal to**

(A) (2, 4) **(B) (4, 4)** **(C) (8, 4)** **(D) None of these**

Sol. **(B)**

Limit exists and denominator $\to 0$

$\therefore$ Numerator must also tend to 0

$\Rightarrow \sqrt{b}-2=0 \Rightarrow b=4$

Using rationalization,

$$\lim_{x\to 0}\frac{\sqrt{ax+4}-2}{x}=\lim_{x\to 0}\frac{ax+4-4}{x\left(\sqrt{ax+4}+2\right)}=1$$

$$\Rightarrow \lim_{x\to 0}\frac{a}{\sqrt{ax+4}+2}=1$$

$$\Rightarrow \frac{a}{2+2}=1\Rightarrow a=4$$

Q.11 $\lim_{x\to\infty}\frac{2x^2-3x+7}{x^2-4x+2}$ **is equal to**

(A) 2 **(B)** $\frac{1}{2}$ **(C)** $\frac{7}{2}$ **(D) Does not exist**

Sol. **(A)**

Divide by x^2 (highest degree)

$$\Rightarrow \lim_{x\to\infty} \frac{2-\frac{3}{x}+\frac{7}{x^2}}{1-\frac{4}{x}+\frac{2}{x^2}} = \frac{2-0+0}{1-0+0} = 2$$

Q.12 $\lim_{x\to\infty} \frac{2x^2-7x-7}{x-4}; x \in N$ **is equal to**

(A) 2 **(B) 0** **(C)** $\frac{7}{4}$ **(D) Does not exist**

Sol. **(D)**

Divide by x in numerator and denominator

$$\lim_{x\to\infty} \frac{2x-7-\frac{7}{x}}{1-\frac{4}{x}} \to \frac{\infty-7}{1-0} \to \infty.$$

∞ is not finite $\therefore$ limit does not exist.

Q.13 $\lim_{x\to\infty} \frac{3x^2-7x+4}{4x^3+3x^2-7x+5}; x \in N$ **is equal to**

(A) $\frac{3}{4}$ **(B)** $\frac{4}{5}$ **(C) 0** **(D) Does not exist**

Sol. **(C)**

Divide by x^2 in numerator and denominator

$$\lim_{x\to\infty} \frac{3-\frac{7}{x}+\frac{4}{x^2}}{4x+3-\frac{7}{x}+\frac{5}{x^2}} = \frac{3}{\infty} = 0$$

Q.14 $\underset{n\to\infty}{\text{Limit}} \frac{1^2+2^2+3^2+\ldots+n^2}{n^3}$ **is equal to**

(A) $\frac{1}{6}$ **(B)** $\frac{1}{3}$ **(C) 0** **(D) Does not exist**

Sol. **(B)**

$$\lim_{n\to\infty} \frac{n(n+1)(2n+1)}{6n^3} = \lim_{n\to\infty} \frac{\frac{n}{n}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)}{6} = \frac{(1)(1+0)(2+0)}{6} = \frac{1}{3}$$

Q.15 $\underset{x\to-\infty}{\text{Limit}}\dfrac{\sqrt{x^2+1}}{3x-6}$ **is equal to**

(A) $\dfrac{1}{3}$ **(B)** $-\dfrac{1}{3}$ **(C)** $\dfrac{1}{6}$ **(D) Does not exist**

Sol. **(B)**

$$\lim_{x\to-\infty}\frac{|x|\sqrt{1+\frac{1}{x^2}}}{x\left(3-\frac{6}{x}\right)}=\lim_{x\to-\infty}\frac{-x\sqrt{1+\frac{1}{x^2}}}{x\left(3-\frac{6}{x}\right)}=\frac{-1\sqrt{1+0}}{3-0}=-\frac{1}{3}$$

Q.16 **Find** $\lim\limits_{x\to-\infty}\dfrac{\sqrt{3x^2-1}-\sqrt{2x^2-1}}{4x+3}$.

Sol. $$\lim_{x\to-\infty}\frac{|x|\sqrt{3-\frac{1}{x^2}}-|x|\sqrt{2-\frac{1}{x^2}}}{x\left(4+\frac{3}{x}\right)}=\frac{-\sqrt{3-0}-\sqrt{2-0}}{4+0}=\frac{-\left(\sqrt{3}+\sqrt{2}\right)}{4}$$

Q.17 **Find the value of** $\underset{x\to-2}{\text{limit}}\left(\dfrac{1}{x+2}-\dfrac{12}{x^3+8}\right)$.

Sol. $$\lim_{x\to-2}\left(\frac{1}{x+2}-\frac{12}{(x+2)(x^2-2x+4)}\right)$$

$$=\lim_{x\to-2}\frac{x^2-2x-8}{(x+2)(x^2-2x+4)}=\lim_{x\to-2}\frac{(x+2)(x-4)}{(x+2)(x^2-2x+4)}$$

$$=\lim_{x\to-2}\frac{x-4}{x^2-2x+4}=\frac{-6}{4+4+4}=-\frac{1}{2}$$

Q.18 **Find the value of** $\underset{x\to\pm\infty}{\text{limit}}\left(\sqrt{x^2-2x-1}-\sqrt{x^2-7x-3}\right)$.

Sol. Using rationalization, we get :

$$\underset{x\to\pm\infty}{\text{limit}}\frac{(x^2-2x-1)-(x^2-7x-3)}{\sqrt{x^2-2x-1}+\sqrt{x^2-7x-3}}=\frac{x\left(5+\frac{2}{x}\right)}{|x|\left(\sqrt{1-\frac{2}{x}-\frac{1}{x^2}}+\sqrt{1-\frac{7}{x}-\frac{3}{x^2}}\right)}$$

$$=\pm\frac{(5+0)}{\sqrt{1-0-0}+\sqrt{1-0-0}}=\pm\frac{5}{2}$$

Q.19 $\lim\limits_{x\to\infty}\left(\sqrt{4x^2+x}-\sqrt{\frac{4x^3}{x+2}}\right)$ **is equal to**

(A) $\frac{4}{9}$ **(B)** $\frac{9}{4}$ **(C)** $\frac{9}{2}$ **(D) None of these**

Sol. **(B)**

$$\lim_{x\to\infty}|x|\left(\sqrt{4+\frac{1}{x}}-\sqrt{\frac{4}{1+\frac{2}{x}}}\right)$$

Now rationalization, we get:

$$\lim_{x\to\infty}\frac{x\left(4+\frac{1}{x}-\frac{4x}{x+2}\right)}{\sqrt{4+\frac{1}{x}}+\sqrt{\frac{4}{1+\frac{2}{x}}}}=\lim_{x\to\infty}\frac{x\left[\frac{(4x+1)(x+2)-4x^2}{x(x+2)}\right]}{\sqrt{4+\frac{1}{x}}+\sqrt{\frac{4}{1+\frac{2}{x}}}}$$

$$=\lim_{x\to\infty}\frac{\frac{9x+2}{x+2}}{\sqrt{4+\frac{1}{x}}+\sqrt{\frac{4}{1+\frac{2}{x}}}}=\lim_{x\to\infty}\frac{9+\frac{2}{x}}{\left(1+\frac{2}{x}\right)\left(\sqrt{4+\frac{1}{x}}+\sqrt{\frac{4}{1+\frac{2}{x}}}\right)}$$

$$=\frac{9}{2+2}=\frac{9}{4}$$

Note :

In case of $f(x)=\frac{\text{Polynomial}}{\text{Polynomial}}$,

where $x\to\pm\infty$; just use the highest degree term in both numerator and denominator to find limit and ignore other terms.

Q.20 $\lim\limits_{x\to\frac{\pi}{2}}\tan^2x\left(\sqrt{2\sin^2x+3\sin x+4}-\sqrt{\sin^2x+6\sin x+2}\right)$ **is equal to**

(A) $\frac{1}{6}$ **(B)** $-\frac{1}{6}$ **(C)** $\frac{1}{12}$ **(D) None of these**

Sol. **(A)**

Rationalizing, we get:

$$\lim_{x\to\frac{\pi}{2}} \frac{\tan^2 x\left(\sin^2 x - 3\sin x + 2\right)}{\sqrt{2\sin^2 x + 3\sin x + 4} + \sqrt{\sin^2 x + 6\sin x + 2}}$$

$$\lim_{x\to\frac{\pi}{2}} \frac{\sin^2 x\left(\sin^2 x - 1\right)\left(\sin^2 x - 2\right)}{\cos^2 x\left(\sqrt{2\sin^2 x + 3\sin x + 4} + \sqrt{\sin^2 x + 6\sin x + 2}\right)}$$

$$= \frac{-1\times(-1)}{\sqrt{2+3+4} + \sqrt{1+6+2}} = \frac{1}{3+3} = \frac{1}{6}$$

Q.21 $\lim_{x\to 0} \frac{ax^3 + bx^2 + cx + d}{x^2}$ **. Limit exists. Find a, b, c, d.**

Sol. It is $\left(\frac{0+0+0+d}{0}\right)$ form.

If $d \neq 0$, then limit does not exist.

So, $d = 0 \Rightarrow \lim_{x\to 0} \frac{ax^3 + bx^2 + cx}{x^2} = \lim_{x\to 0} \frac{ax^2 + bx + c}{x}$

Again, c must be 0 for limit to exist.

$\therefore \lim_{x\to 0} \frac{ax^2 + bx}{x} = \lim_{x\to 0} ax + b = b$

$\therefore$ a, b $\in$ R and c, d = 0

Q.22 **If** $\lim_{x\to 0} \frac{f(x)}{x^4} = 3$ **where f is a four degree polynomial, then the value of f(1) is**

(A) 3 **(B)** $\frac{1}{3}$ **(C) 9** **(D) None of these**

Sol. **(A)**

Let $f(x) = ax^4 + bx^3 + cx^2 + dx + e$

$\therefore \lim_{x\to 0} \frac{f(x)}{x^4} = \lim_{x\to 0}\left(a + \frac{b}{x} + \frac{c}{x^2} + \frac{d}{x^3} + \frac{e}{x^4}\right)$

So, for limit to exist, b = c = d = e = 0

$\lim_{x\to 0} a = 3 \Rightarrow a = 3$

$\Rightarrow f(1) = 3 \times 1 + 0 + 0 + 0 + 0 = 3$

Q.23 $\lim\limits_{x\to\frac{\pi}{2}} \dfrac{e^{\tan x}-1}{e^{\tan x}+1}$

(A) 1 **(B) 0** **(C) −1** **(D) None of these**

Sol. **(D)**

As $x \to \dfrac{\pi}{2} \to \tan x \to \pm\infty \to \left(\text{discontinuity at } x = \dfrac{\pi}{2}\right)$

So, check for LHL and RHL separately.

$$\Rightarrow x \to \frac{\pi^-}{2} \Rightarrow \lim_{x\to\frac{\pi^-}{2}} \frac{e^{\tan x}-1}{e^{\tan x}+1} = \lim_{x\to\frac{\pi^-}{2}} \frac{e^{\tan x}\left(1-\frac{1}{e^{\tan x}}\right)}{e^{\tan x}\left(1+\frac{1}{e^{\tan x}}\right)} = \frac{1-0}{1+0} = 1 = \text{LHL}$$

$$x \to \frac{\pi^+}{2} \Rightarrow \tan x \to -\infty \Rightarrow e^{\tan x} \to 0$$

$$\therefore \lim_{x\to\frac{\pi^+}{2}} \frac{e^{\tan x}-1}{e^{\tan x}+1} = \frac{0-1}{0+1} = -1 = \text{RHL}$$

$\therefore$ LHL $\neq$ RHL

$\therefore$ Limit does not exist.

Q.24 **Evaluate** $\lim\limits_{n\to\infty} \prod\limits_{r=3}^{n} \dfrac{r^3-8}{r^3+8}$ **, where** $\prod$ **represents product of function.**

(A) $\dfrac{2}{3}$ **(B)** $\dfrac{2}{5}$ **(C)** $\dfrac{1}{3}$ **(D)** $\dfrac{2}{7}$

Sol. **(D)**

$$P = \lim_{r\to\infty} \prod_{r=3}^{n} \frac{r^3-8}{r^3+8}$$

$$= \lim_{n\to\infty} \prod_{r=3}^{r} \left(\frac{r-2}{r+2}\right)\left(\frac{r^2+2r+4}{r^2-2r+4}\right)$$

$$= \lim_{n\to\infty} \prod_{r=3}^{r} \left(\frac{r-2}{r+2}\right) \prod_{r=3}^{n} \left(\frac{r^2+2r+4}{r^2-2r+4}\right)$$

$$= \lim_{n\to\infty} \left\{\frac{1}{5}\frac{2}{6}\frac{3}{7}\frac{4}{8}\frac{5}{9} \cdots \frac{(n-5)}{(n-1)}\frac{(n-4)}{(n)}\frac{(n-3)}{(n+1)}\frac{(n-2)}{(n+2)}\right\} \times$$

$$\left\{\frac{19}{7} \times \frac{28}{12} \times \frac{39}{19} \cdots \frac{(n^2-2n+4)}{(n^2-6n+12)} \times \frac{(n^2+3)}{(n^2-4n+7)} \times \frac{(n^2+2n+4)}{(n^2-2n+4)}\right\}$$

$$= \lim_{n\to\infty}\left\{\frac{1\cdot 2\cdot 3\cdot 4}{(n-1)n(n+1)(n+2)}\times\frac{(n^2+3)(n^2+2n+4)}{7\times 12}\right\}$$

$$= \frac{2}{7}\lim_{n\to\infty}\left\{\frac{(n^2+3)(n^2+2n+4)}{(n-1)n(n+1)(n-2)}\right\}$$

$$= \frac{2}{7}\lim_{n\to\infty}\left\{\frac{\left(1+\frac{3}{n^2}\right)\left(1+\frac{2}{n}+\frac{4}{n^2}\right)}{\left(1-\frac{1}{n}\right)1\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)}\right\} = \frac{2}{7}\frac{(1+0)(1+0+0)}{(1-0)(1+0)(1+0)} = \frac{2}{7}$$

Hence $P = \frac{2}{7}$

Q.25 **Evaluate** $\lim_{x\to 0}\frac{\sqrt{1+x}-1}{x}$.

Sol. When $x = 0$, the expression $\frac{\sqrt{1+x}-1}{x}$ takes the form $\frac{0}{0}$,

Rationalizing the numerator, we have

$$\lim_{x\to 0}\frac{\sqrt{1+x}-1}{x} = \lim_{x\to 0}\frac{\left(\sqrt{1+x}-1\right)\left(\sqrt{1+x}+1\right)}{x\left(\sqrt{1+x}+1\right)}$$

$$= \lim_{x\to 0}\frac{1+x-1}{\left(\sqrt{1+x}+1\right)x} = \lim_{x\to 0}\frac{1}{\sqrt{1+x}+1} = \frac{1}{2}$$

Q.26 **Evaluate** $\lim_{x\to 9}\frac{3-\sqrt{x}}{4-\sqrt{2x-2}}$.

Sol. $\lim_{x\to 9}\frac{\left(3-\sqrt{x}\right)}{4-\sqrt{2x-2}}\times\frac{\left(3+\sqrt{x}\right)}{\left(3+\sqrt{x}\right)}\times\frac{4+\sqrt{2x-2}}{4+\sqrt{2x-2}}\left(\frac{0}{0}\right)$

$$= \lim_{x\to 9}\frac{(9-x)}{16-(2x-2)}\times\frac{\left(4+\sqrt{2x-2}\right)}{\left(3+\sqrt{x}\right)}$$

$$= \lim_{x\to 9}\frac{(9-x)}{2(9-x)}\times\frac{\left(4+\sqrt{2x-2}\right)}{\left(3+\sqrt{x}\right)}$$

$$= \lim_{x\to 9}\frac{1}{2}\times\frac{\left(4+\sqrt{2x-2}\right)}{3+\sqrt{x}} = \frac{1}{2}\times\frac{4+\sqrt{16}}{3+\sqrt{9}} = \frac{2}{3}$$

Q.27 **Evaluate** $\lim_{x\to\infty}\sqrt[3]{x^3+3x^2}-\sqrt{x^2-2x}$.

Sol. $\lim_{x\to\infty}\sqrt[3]{x^3+3x^2}-\sqrt{x^2-2x}$

$$=\lim_{x\to\infty}\left[x^3\left(1+\frac{3}{x}\right)\right]^{\frac{1}{3}}-\left[x^2\left(1-\frac{2}{x}\right)\right]^{\frac{1}{2}}$$

$$=\lim_{x\to\infty}x\left[\left(1+\frac{3}{x}\right)^{\frac{1}{3}}-\left(1-\frac{2}{x}\right)^{\frac{1}{2}}\right]$$

$$=\lim_{x\to\infty}\frac{\left(1+\frac{3}{x}\right)^{\frac{1}{3}}-1-\left(1-\frac{2}{x}\right)^{\frac{1}{2}}+1}{\left(\frac{1}{x}\right)}$$

$$=\lim_{x\to\infty}\frac{3\left[\left(1+\frac{3}{x}\right)^{\frac{1}{3}}-1^{\frac{1}{3}}\right]}{\frac{3}{x}}+\frac{\left[\left(1-\frac{2}{x}\right)^{\frac{1}{2}}-1\right]2}{\left(-\frac{2}{x}\right)}$$

$$=\lim_{x\to\infty}\frac{3\left[\left(1+\frac{3}{x}\right)^{\frac{1}{3}}-1^{\frac{1}{3}}\right]}{\left[\left(1+\frac{3}{x}\right)-1\right]}+\frac{\left[\left(1-\frac{2}{x}\right)^{\frac{1}{2}}-1^{\frac{1}{2}}\right]2}{\left[\left(1-\frac{2}{x}\right)-1\right]}$$

$$=\lim_{t\to1}\frac{3\left(t^{\frac{1}{3}}-1^{\frac{1}{3}}\right)}{(t-1)}+\lim_{p\to1}\frac{\left(p^{\frac{1}{2}}-1^{\frac{1}{2}}\right)}{p-1}\times2 \qquad \left(\text{Let } \left(1+\frac{3}{x}\right)=t \ \& \ \left(1-\frac{2}{x}\right)=p\right)$$

$$=3\times\left[\frac{1}{3}(1)^{\frac{-2}{3}}\right]+2\left(\frac{1}{2}(1)^{\frac{-1}{2}}\right)=1+1=2$$

Q.28 **Evaluate** $\lim_{x\to\infty}\left((x+1)(x+2)(x+3)\right)^{\frac{1}{3}}-x$.

Sol. $\lim_{x\to\infty}\left((x+1)(x+2)(x+3)\right)^{\frac{1}{3}}-x$

$$=\lim_{y\to0}\frac{\left[(1+y)(1+2y)(1+3y)\right]^{\frac{1}{3}}}{y}-\frac{1}{y} \qquad \left(\text{Put } x=\frac{1}{y}\right)$$

$$= \lim_{y\to 0} \frac{\left(6y^3 + 11y^2 + 6y + 1\right)^{\frac{1}{3}} - 1}{y}$$

$$= \lim_{y\to 0} \frac{1 + \frac{1}{3}(6y) + \text{Higher degree terms} - 1}{y} = 2$$

Q.29 $\lim\limits_{x\to 0}\left(1^{\text{cosec}^2 x} + 2^{\text{cosec}^2 x} + 3^{\text{cosec}^2 x} + \ldots + 100^{\text{cosec}^2 x}\right)^{\sin^2 x}$

Sol. $\lim\limits_{x\to 0}\left(1^{\text{cosec}^2 x} + 2^{\text{cosec}^2 x} + 3^{\text{cosec}^2 x} + \ldots + 100^{\text{cosec}^2 x}\right)^{\sin^2 x}$

$$= 100 \lim_{x\to 0}\left(\left(\frac{1}{100}\right)^{\text{cosec}^2 x} + \left(\frac{2}{100}\right)^{\text{cosec}^2 x} + \ldots + \left(\frac{99}{100}\right)^{\text{cosec}^2 x} + 1\right)^{\sin^2 x}$$

$$= 100\,(0 + 0 + \ldots + 0 + 1)^0 = 100$$

SANDWICH THEOREM OR SQUEEZE PLAY THEOREM FOR EVALUATING LIMITS :

General :

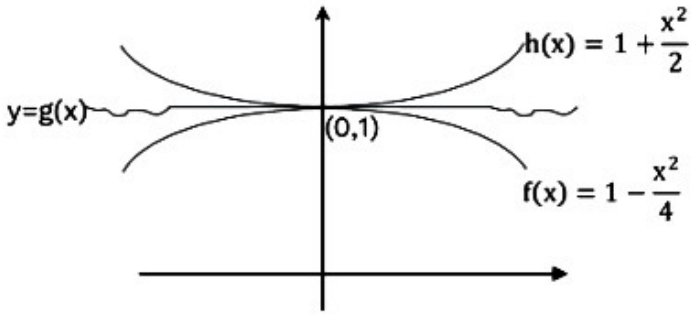
The squeeze principle is used on limit problems where the usual algebraic methods (factorisation or algebraic manipulation etc.) are not effective. However, it requires to "squeeze" our problem in between two other simpler function whose limits can be easily computed and equal. Use of squeeze principle requires accurate analysis, in-depth algebra skills careful use of inequalities.

Statement :

If f, g and h are 3 functions such that $f(x) \le g(x) \le h(x)$ for all x in some interval containing the point x = c, and if

$$\lim_{x\to c} f(x) = \lim_{x\to c} h(x) = L \Rightarrow \lim_{x\to c} g(x) = L$$

From the figure note that $\lim\limits_{x\to 0} g(x) = 1$



Note :

The quantity may be a finite number, $+\infty$ or $-\infty$.
Similarly, L may be finite number, $+\infty$ or $-\infty$.

Q.1 **Evaluate $\lim\limits_{x\to\infty} \dfrac{x + 7\sin x}{-2x + 13}$ using Sandwich theorem.**

Sol. We know that, $-1 \le \sin x \le 1$ for all x.

$\Rightarrow \; -7 \le 7 \sin x \le 7$

$\Rightarrow \; x - 7 \le x + 7 \sin x \le x + 7$

Dividing throughout by $-2x + 13$, we get

$\dfrac{x - 7}{-2x+13} \ge \dfrac{x+7\sin x}{-2x+13} \ge \dfrac{x + 7}{-2x+13}$ for all x that are large.

Now, $\lim\limits_{x\to\infty} \dfrac{x - 7}{-2x+13} = \lim\limits_{x\to\infty} \dfrac{1 - \frac{7}{x}}{-2 + \frac{13}{x}} = \dfrac{1-0}{-2+0} = -\dfrac{1}{2}$

and $\lim\limits_{x\to\infty} \dfrac{x + 7}{-2x+13} = \lim\limits_{x\to\infty} \dfrac{1 + \frac{7}{x}}{-2 + \frac{13}{x}} = \dfrac{1+0}{-2+0} = -\dfrac{1}{2}$

$\therefore \; \lim\limits_{x\to\infty} \dfrac{x + 7\sin x}{-2x + 13} = \dfrac{-1}{2}$

Q.2 **If $[\cdot]$ denotes the greatest integer function, then find the value of**

$\lim\limits_{x\to\infty} \dfrac{[x]+[2x]+\ldots+[nx]}{n^2}$.

Sol. $nx - 1 < [nx] \le nx$

Putting n=1, 2, 3, ..., n and adding them,

$x\sum n - n < \sum [nx] \le x\sum n$

$\therefore \; x\dfrac{\sum n}{n^2} - \dfrac{1}{n} < \dfrac{\sum [nx]}{n^2} \le x.\dfrac{\sum n}{n^2}$ ...(1)

Now, $\lim\limits_{n\to\infty} \left\{x.\dfrac{\sum n}{n^2} - \dfrac{1}{n}\right\} = x.\lim\limits_{n\to} \dfrac{\sum n}{n^2} - \lim\limits_{x\to\infty} \dfrac{1}{n} = \dfrac{x}{2}$

$$\lim_{n\to\infty}\left\{x.\frac{\Sigma n}{n^2}\right\} = x\lim_{n\to\infty}\frac{\Sigma n}{n^2} = \frac{x}{2}$$

As the two limits are equal, by equation (1),

$$\lim_{n\to\infty}\frac{\Sigma[nx]}{n^2} = \frac{x}{2}$$

Q.3 **Evaluate** $\lim_{n\to\infty}\frac{1}{1+n^2}+\frac{2}{2+n^2}+\ldots+\frac{n}{n+n^2}$.

Sol. $P_n = \frac{1}{1+n^2}+\frac{2}{2+n^2}+\ldots+\frac{n}{n+n^2}$

Now, $P_n < \frac{1}{1+n^2}+\frac{2}{1+n^2}+\ldots+\frac{n}{1+n^2} = \frac{1+2+\ldots+n}{1+n^2} = \frac{n(n+1)}{2(1+n^2)}$

Also, $P_n > \frac{1}{n+n^2}+\frac{2}{n+n^2}+\frac{3}{n+n^2}+\ldots+\frac{n}{n+n^2} = \frac{1+2+\ldots+n}{(n+n^2)} = \frac{n(n+1)}{2(n+n^2)}$

Thus, $\frac{n(n+1)}{2(n+n^2)} < P_n < \frac{n(n+1)}{2(1+n^2)}$

$$\Rightarrow \lim_{n\to\infty}\frac{n(n+1)}{2(n+n^2)} < \lim_{n\to\infty}P_n < \lim_{n\to\infty}\frac{n(n+1)}{2(1+n^2)}$$

$$\Rightarrow \lim_{n\to\infty}\frac{1\left(1+\frac{1}{n}\right)}{2\left(\frac{1}{n}+1\right)} < \lim_{n\to\infty}P_n < \lim_{n\to\infty}\frac{1\left(1+\frac{1}{n}\right)}{2\left(\frac{1}{n^2}+1\right)}$$

$$\Rightarrow \frac{1}{2} < \lim_{n\to\infty}P_n < \frac{1}{2} \Rightarrow \lim_{n\to\infty}P_n = \frac{1}{2}$$

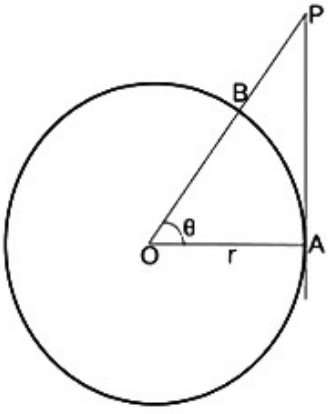
EVALUATION OF TRIGONOMETRIC LIMITS :

If θ is small and is measured in radians then

(i) $\lim_{\theta\to 0}\frac{\sin\theta}{\theta} = 1$

Proof :

Consider a circle of radius r. Let O be the centre of the circle such that $\angle AOB = \theta$ where θ is measured in radians and it is very small. Suppose the tangent at A meets OB produced at P. From figure, we have



Area of ΔOAB < Area of sector OAB < Area of ΔOAP

$\Rightarrow \frac{1}{2}OA\times OB\sin\theta < \frac{1}{2}(OA)^2\theta < \frac{1}{2}OA\times AP$

$\Rightarrow \frac{1}{2}r^2\sin\theta < \frac{1}{2}r^2\theta < \frac{1}{2}r^2\tan\theta$ [In ΔOAP, $AP=OA\tan\theta$]

$\Rightarrow \sin\theta < \theta < \tan\theta \quad \Rightarrow 1 > \frac{\sin\theta}{\theta} > \cos\theta$

$\Rightarrow 1 > \lim_{\theta\to0}\frac{\sin\theta}{\theta} > \lim_{\theta\to0}\cos\theta$ or, $\lim_{\theta\to0}\cos\theta < \lim_{\theta\to0}\frac{\sin\theta}{\theta} < 1$

$\Rightarrow 1 < \lim_{\theta\to0}\frac{\sin\theta}{\theta} < 1 \Rightarrow \lim_{\theta\to0}\frac{\sin\theta}{\theta} = 1$ (By Sandwich Theorem)

(ii) $\lim_{\theta\to0}\frac{\tan\theta}{\theta} = 1$

We have $\lim_{\theta\to0}\frac{\tan\theta}{\theta} = \lim_{\theta\to0}\frac{\sin\theta}{\theta}\frac{1}{\cos\theta} = \lim_{\theta\to0}\frac{\sin\theta}{\theta}.\lim_{\theta\to0}\frac{1}{\cos\theta} = 1$

(iii) $\lim_{\theta\to a}\frac{\sin(\theta-a)}{\theta-a} = 1$

We have $\lim_{\theta\to a}\frac{\sin(\theta-a)}{\theta-a} = \lim_{h\to0}\frac{\sin(a+h-a)}{(a+h-a)} = \lim_{h\to0}\frac{\sinh}{h} = 1$

(iv) $\lim_{\theta\to a}\frac{\tan(\theta-a)}{\theta-a} = 1$

(v) $\lim_{\theta\to0}\frac{\sin^{-1}\theta}{\theta} = 1$

(vi) $\lim_{\theta\to0}\frac{\tan^{-1}\theta}{\theta} = 1$

Note :

Let $[\cdot]$ denotes greatest integer function

(i) $\lim_{x\to0}\left[\frac{\sin x}{x}\right] = 0$ **(ii)** $\left[\lim_{x\to0}\frac{\sin x}{x}\right] = 1$

(iii) $\lim_{x\to0}\left[\frac{\tan x}{x}\right] = 1$ **(iv)** $\lim_{x\to0}\left[\frac{\sin^{-1}x}{x}\right] = 1$

(v) $\left[\lim_{x\to0}\frac{\sin^{-1}x}{x}\right] = 1$

Q.1 **Evaluate** $\lim_{x\to 0}\frac{\sin 2x}{5x}$.

Sol. We have $\lim_{x\to 0}\frac{\sin 2x}{5x}=\lim_{x\to 0}\left(\frac{2}{5}\right)\left(\frac{\sin 2x}{2x}\right)=\frac{2}{5}\lim_{x\to 0}\frac{\sin 2x}{2x}=\frac{2}{5}(1)=\frac{2}{5}$

$\left[\because \lim_{x\to 0}\frac{\sin 2x}{2x}=1\right]$

Q.2 **Evaluate** $\lim_{x\to 0}\frac{1-\cos x}{x^2}$.

Sol. $\lim_{x\to 0}\frac{1-\cos x}{x^2}=\lim_{x\to 0}\frac{2\sin^2\left(\frac{x}{2}\right)}{4\left(\frac{x}{2}\right)^2}=\frac{1}{2}\times 1=\frac{1}{2}$

Note :

$\lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2}$ is a standard limit and should be remembered.

Q.3 **Evaluate** $\lim_{x\to 0}\frac{1-\cos 5x}{3x^2}$.

Sol. $\lim_{x\to 0}\frac{1-\cos 5x}{3x^2}=\lim_{x\to 0}\left(\frac{25}{3}\right)\frac{1-\cos 5x}{(5x)^2}=\frac{25}{3}\lim_{x\to 0}\frac{1-\cos 5x}{(5x)^2}$

$=\frac{25}{3}\times\frac{1}{2}=\frac{25}{6}$

Q.4 **Evaluate** $\lim_{x\to 1}\frac{x^3+x^2-2}{\sin(x-1)}$.

Sol. $\lim_{x\to 1}\frac{x^3+x^2-2}{\sin(x-1)}=\lim_{x\to 1}\frac{(x^3-1)+(x^2-1)}{\sin(x-1)}$

$\lim_{x\to 1}\frac{(x-1)\left[(x^2+1+x)+(x+1)\right]}{\sin(x-1)}=\lim_{x\to 1}\frac{(x-1)}{\sin(x-1)}\times(x^2+2x+2)=1\times(1+2+2)=5$

Q.5 **Evaluate** $\lim_{x\to 0}\frac{\sqrt{1+\tan x}-\sqrt{1+\sin x}}{x^3}$.

Sol. $\lim_{x\to 0}\frac{\sqrt{1+\tan x}-\sqrt{1+\sin x}}{x^3}\times\frac{\left(\sqrt{1+\tan x}+\sqrt{1+\sin x}\right)}{\left(\sqrt{1+\tan x}+\sqrt{1+\sin x}\right)}$

$\lim_{x\to 0}\frac{\tan x-\sin x}{x^3\left(\sqrt{1+\tan x}+\sqrt{1+\sin x}\right)}$

$\lim_{x\to 0}\frac{\sin x(1-\cos x)}{x^3\cos x}\times\frac{1}{\left(\sqrt{1+\tan x}+\sqrt{1+\sin x}\right)}$

$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)\times\lim_{x\to 0}\frac{(1-\cos x)}{x^2}\times\lim_{x\to 0}\frac{1}{\cos x\left(\sqrt{1+\tan x}+\sqrt{1+\sin x}\right)}$

$=\lim_{x\to 0}\frac{\sin x}{x}\times\lim_{x\to 0}\frac{2\sin^2\left(\frac{x}{2}\right)}{4\times\left(\frac{x^2}{4}\right)}\times\lim_{x\to 0}\frac{1}{\cos x\left(\sqrt{1+\tan x}+\sqrt{1+\sin x}\right)}$

$=1\times\frac{1}{2}\times 1\times\frac{1}{1(1+1)}=\frac{1}{4}$

Q.6 **Evaluate** $\lim_{x\to 1}(1-x)\tan\frac{\pi x}{2}$.

Sol. $\lim_{x\to 1}(1-x)\frac{\sin\left(\frac{\pi x}{2}\right)}{\cos\left(\frac{\pi x}{2}\right)}=\lim_{x\to 1}(1-x)\frac{\sin\left(\frac{\pi x}{2}\right)}{\sin\frac{\pi}{2}(1-x)}$

$=\lim_{x\to 1}\frac{\frac{\pi}{2}(1-x)}{\sin\left(\frac{\pi}{2}(1-x)\right)}\times\frac{1}{\left(\frac{\pi}{2}\right)}\times\sin\left(\frac{\pi x}{2}\right)$

$=1\times\frac{2}{\pi}\times 1=\frac{2}{\pi}$

Q.7 **Evaluate** $\lim_{x\to 0}\frac{1-\cos\left(1-\cos x\right)}{\sin^4 x}$.

Sol. $\lim_{x\to 0}\frac{2\sin^2\left(\frac{1-\cos x}{2}\right)}{\sin^4 x}=\lim_{x\to 0}\frac{2\sin^2\left(\sin^2\frac{x}{2}\right)}{\sin^4 x}$

$$=\lim_{x\to 0}\frac{2\sin^2\left(\sin^2\left(\frac{x}{2}\right)\right)\times\left[\sin^2\left(\frac{x}{2}\right)\right]^2\times\left(\frac{x}{2}\right)^4}{\left[\sin^2\left(\frac{x}{2}\right)\right]^2\times\left(\frac{x}{2}\right)^4\times\sin^4 x}$$

$$=2\times 1\times 1\times\frac{1}{16}=\frac{1}{8}$$

Q.8 **Evaluate** $\lim_{x\to 0}\frac{1-\cos x\sqrt{\cos 2x}}{\tan^2 x}$.

Sol. $\lim_{x\to 0}\frac{1-\cos x\sqrt{\cos 2x}}{\tan^2 x}\times\frac{\left(1+\cos x\sqrt{\cos 2x}\right)}{\left(1+\cos x\sqrt{\cos 2x}\right)}$

$$=\lim_{x\to 0}\frac{1-\left(\cos^2 x\right)\left(2\cos^2 x-1\right)}{\left(\frac{\tan^2 x}{x^2}\times x^2\right)}\times\frac{1}{\left(1+\cos x\sqrt{\cos 2x}\right)}$$

$$=\lim_{x\to 0}\frac{\sin^2 x}{x^2}\frac{\left(1+2\cos^2 x\right)}{1+\cos x\sqrt{\cos 2x}}\times\frac{1}{\left(\frac{\tan^2 x}{x^2}\right)}$$

$$=1\times\frac{1+2}{1+1}\times\frac{1}{1}=\frac{3}{2}$$

Q.9 **Evaluate** $\lim_{x\to 0}\frac{1}{x}\sin^{-1}\left(\frac{2x}{1+x^2}\right)$.

Sol. We know that, $\sin^{-1}\left(\frac{2x}{1+x^2}\right)=2\tan^{-1}x$, for $-1\le x\le 1$

$$\Rightarrow\lim_{x\to 0}\frac{1}{x}\sin^{-1}\left(\frac{2x}{1+x^2}\right)=\lim_{x\to 0}\frac{2\tan^{-1}x}{x}=2$$

Q.10 **Evaluate** $\lim_{x\to\infty} x\left(\tan^{-1}\frac{x+1}{x+4}-\frac{\pi}{4}\right)$

Sol. We have $\lim_{x\to\infty} x\left(\tan^{-1}\frac{x+1}{x+4}-\frac{\pi}{4}\right)$

$$=\lim_{x\to\infty} x\left(\tan^{-1}\frac{x+1}{x+4}-\tan^{-1}1\right)=\lim_{x\to\infty} x\tan^{-1}\left(\frac{\frac{x+1}{x+4}-1}{1+\frac{x+1}{x+4}}\right)$$

$$=\lim_{x\to\infty} x\tan^{-1}\left(\frac{-3}{2x+5}\right)=\lim_{x\to\infty}\left(\frac{\tan^{-1}\left(\frac{-3}{2x+5}\right)}{\frac{-3}{2x+5}}\right)\left(\frac{-3}{2x+5}\right)x$$

$$=\lim_{x\to\infty}\left(\frac{\tan^{-1}\left(\frac{-3}{2x+5}\right)}{\frac{-3}{2x+5}}\right)\lim_{x\to\infty}\left(\frac{-3x}{2x+5}\right)=1\times\lim_{x\to\infty}\left(\frac{-3}{2+\frac{5}{x}}\right)=1\times\frac{-3}{2}=-\frac{3}{2}$$

Q.11 **Evaluate** $\lim_{x\to\frac{\pi}{2}}\frac{1+\cos 2x}{(\pi-2x)^2}$

Sol. We have, $\lim_{x\to\frac{\pi}{2}}\frac{1+\cos 2x}{(\pi-2x)^2}\left(\frac{0}{0}\text{ form}\right)$

$$=\lim_{h\to 0}\frac{1+\cos 2\left(\frac{\pi}{2}+h\right)}{\left[\pi-2\left(\frac{\pi}{2}+h\right)\right]^2}=\lim_{h\to 0}\frac{1+\cos(\pi+2h)}{4h^2}=\lim_{h\to 0}\frac{1-\cos 2h}{4h^2}$$

$$=\lim_{h\to 0}\frac{2\sin^2 h}{4h^2}=\frac{2}{4}\left(\lim_{h\to 0}\frac{\sinh}{h}\right)^2=\frac{1}{2}$$

Q.12 $\lim_{x\to\frac{\pi}{4}}\frac{\sqrt{2}\cos x-1}{\cot x-1}$

Sol. $\lim_{x\to\frac{\pi}{4}}\frac{(\sqrt{2}\cos x-1)\sin x}{\cos x-\sin x}$

$$\lim_{x\to\frac{\pi}{4}}\frac{(\sqrt{2}\cos x-1)(\sqrt{2}\cos x+1)}{(\sqrt{2}\cos x+1)(\cos x-\sin x)}\times\frac{(\cos x+\sin x)}{(\cos x+\sin x)}.\sin x$$

$$= \lim_{x \to \frac{\pi}{4}} \frac{(2\cos^2 x - 1)}{\cos^2 x - \sin^2 x} \times \frac{(\cos x + \sin x).\sin x}{(\sqrt{2}\cos x + 1)}$$

$$= \lim_{x \to \frac{\pi}{4}} (1) \times \frac{(\cos x + \sin x).(\sin x)}{(\sqrt{2}\cos x + 1)} = \frac{\left(\frac{1}{\sqrt{2}} \times 2\right).\frac{1}{\sqrt{2}}}{\left(\sqrt{2} \times \frac{1}{\sqrt{2}} + 1\right)} = \frac{1}{2}$$

Q.13 $\lim_{x \to 0} \frac{\cot 2x - \operatorname{cosec} 2x}{x}$

Sol. $\lim_{x \to 0} \frac{\frac{\cos 2x}{\sin 2x} - \frac{1}{\sin 2x}}{x}$

$$= \lim_{x \to 0} \frac{-(1 - \cos 2x)}{(\sin 2x) \times x} = \lim_{x \to 0} \frac{-(2\sin^2 x)}{(2\sin x \cos x)x}$$

$$= \lim_{x \to 0} -\left(\frac{\sin x}{x}\right) \times \frac{1}{\cos x} = (-1) \times 1 = -1$$

Q.14 $\lim_{x \to 0} \frac{\tan 2x - x}{3x - \sin x}$

Sol. $\lim_{x \to 0} \frac{\left(\frac{\tan 2x - x}{2x}\right)}{\left(\frac{3x - \sin x}{2x}\right)} = \lim_{x \to 0} \frac{\frac{\tan 2x}{2x} - \frac{1}{2}}{\frac{3}{2} - \frac{1}{2}\left(\frac{\sin x}{x}\right)}$

$$= \frac{1 - \frac{1}{2}}{\frac{3}{2} - \frac{1}{2}(1)} = \frac{\frac{1}{2}}{1} = \frac{1}{2}$$

EVALUATION OF EXPONENTIAL AND LOGARITHMIC LIMITS :

In order to evaluate these type of limit, we use the following standard results.

1. $\lim_{x \to 0} \frac{a^x - 1}{x} = \log_e a$

Proof :

$$\lim_{x\to 0}\frac{a^x-1}{x}=\lim_{x\to 0}\frac{\left(1+\frac{x(\log a)}{1!}+\frac{x^2(\log a)^2}{2!}+\ldots\right)-1}{x}=\lim_{x\to 0}\left(\frac{\log a}{1!}+\frac{x(\log a)^2}{2!}+\ldots\right)=\log_e a$$

2. $\lim_{x\to 0}\frac{e^x-1}{x}=1$ (replace a by e in the above proof)

3. $\lim_{x\to 0}\frac{\log(1+x)}{x}=1$

Proof :

$$\lim_{x\to 0}\frac{\log(1+x)}{x}=\lim_{x\to 0}\frac{x-\frac{x^2}{2}+\frac{x^3}{3}-\ldots}{x}=\lim_{x\to 0}\left(1-\frac{x}{2}+\frac{x^2}{3}\ldots\right)=1$$

Q.1 $\lim_{x\to 0}\frac{e^{\tan x}-e^x}{\tan x-x}$

Sol. $\lim_{x\to 0}\frac{\left(e^{\tan x}-e^x\right)}{(\tan x-x)}=\lim_{x\to 0}\frac{e^x\left(e^{\tan x-x}-1\right)}{(\tan x-x)}$

$= e^0 \times 1 = 1$

Q.2 $\lim_{x\to 0}\frac{e^{x^2}-\cos x}{x^2}$

Sol. $\lim_{x\to 0}\frac{\left(e^{x^2}-1\right)+(1-\cos x)}{x^2}=\lim_{x\to 0}\frac{e^{x^2}-1}{x^2}+\lim_{x\to 0}\frac{2\sin^2\frac{x}{2}}{x^2}$

$$=1+\lim_{x\to 0}2\left[\frac{\sin^2\left(\frac{x}{2}\right)}{4\times\left(\frac{x}{2}\right)^2}\right]$$

$$=1+\frac{2}{4}\times 1=\frac{3}{2}$$

Q.3 $\lim_{x\to 0}\frac{e^{\sin 2x}-e^{\sin x}}{x}$

Sol. $\lim_{x\to 0}\frac{e^{\sin 2x}-e^{\sin x}}{x}=\lim_{x\to 0}\frac{e^{\sin 2x}-1-e^{\sin x}+1}{x}$

$$\lim_{x\to 0}\frac{e^{\sin 2x}-1}{x}-\frac{e^{\sin x}-1}{x}$$

$$=\lim_{x\to 0}\left(\frac{e^{\sin 2x}-1}{\sin 2x}\cdot\frac{\sin 2x}{2x}\cdot 2-\frac{e^{\sin x}-1}{\sin x}\cdot\frac{\sin x}{x}\right)$$

$$=1\times 1\times 2-1\times 1$$

$$=1$$

Q.4 $\lim_{x\to 0}\frac{e^{x}+e^{-x}-2}{x^{2}}$

Sol. $\lim_{x\to 0}\frac{e^{x}+\frac{1}{e^{x}}-2}{x^{2}}=\lim_{x\to 0}\frac{\left(e^{x}-1\right)^{2}}{e^{x}\left(x^{2}\right)}$

$$\lim_{x\to 0}\left(\frac{e^{x}-1}{x}\right)^{2}\times\frac{1}{e^{x}}=1\times\frac{1}{1}=1$$

Q.5 $\lim_{x\to a}\frac{\log(x-a)}{\log\left(e^{x}-e^{a}\right)}$

Sol. $\lim_{x\to a}\frac{\log(x-a)}{\log\left[e^{a}\left(e^{x-a}-1\right)\right]}=\lim_{x\to a}\frac{\log(x-a)}{\log e^{a}+\log\left[\frac{e^{x-a}-1}{x-a}\times(x-a)\right]}$

$$=\lim_{x\to a}\frac{\log(x-a)}{a+\log(x-a)}$$

$$=\lim_{x\to a}\frac{1}{\frac{a}{\log(x-a)}+1}=\frac{1}{0+1}=1$$

Q.6 $\lim\limits_{x\to\infty}\dfrac{e^{\frac{1}{x^2}}-1}{2\arctan x^2-\pi}$

Sol. $\lim\limits_{x\to\infty}\dfrac{e^{\frac{1}{x^2}}-1}{2\left(\frac{\pi}{2}-\cot^{-1}x^2\right)-\pi}=\lim\limits_{x\to\infty}\dfrac{\left(e^{\frac{1}{x^2}}-1\right)}{\pi-2\cot^{-1}x^2-\pi}$

put $x=\dfrac{1}{y}$

$$\lim_{y\to0}\frac{\left(e^{y^2}-1\right)}{-2\cot^{-1}\frac{1}{y^2}}=\frac{-1}{2}\lim_{y\to0}\frac{\left(e^{y^2}-1\right)}{\left(\frac{\tan^{-1}y^2}{y^2}\right)\times y^2}$$

$$=\frac{-1}{2}\lim_{y\to0}\frac{\left(\frac{e^{y^2}-1}{y^2}\right)}{\left(\frac{\tan^{-1}y^2}{y^2}\right)}=\frac{-1}{2}\times\frac{1}{1}=\frac{-1}{2}$$

Q.7 $\lim\limits_{x\to2}\dfrac{\sin\left(e^{x-2}-1\right)}{\log\left(x-1\right)}$

Sol. $\lim\limits_{x\to2}\dfrac{\sin\left(e^{x-2}-1\right)}{\log\left(x-1\right)}$

$$=\lim_{x\to2}\frac{\sin\left[\frac{e^{(x-2)}-1}{(x-2)}(x-2)\right]}{\frac{\log\left[1+(x-2)\right]}{(x-2)}\times(x-2)}$$

$$=\lim_{x\to2}\frac{\sin\left(1\times(x-2)\right)}{1\times(x-2)}=1$$

Q.8 $\lim\limits_{x\to0}\dfrac{\cos\left(xe^{x}\right)-\cos\left(xe^{-x}\right)}{x^3}$

Sol. $\lim\limits_{x\to0}\dfrac{2\sin\left[\frac{x}{2}\left(e^{x}+e^{-x}\right)\right]\sin\left[\frac{x}{2}\left(e^{-x}-e^{x}\right)\right]\times\frac{x^2}{4}\left(e^{x}+e^{-x}\right)\left(e^{-x}-e^{x}\right)}{x^3\times\left[\frac{x}{2}\left(e^{x}+e^{-x}\right)\right]\times\left[\frac{x}{2}\left(e^{-x}-e^{x}\right)\right]}$

$$= \lim_{x\to 0} \frac{\sin\left[\frac{x}{2}\left(e^{x}+e^{-x}\right)\right]}{\frac{x}{2}\left(e^{x}+e^{-x}\right)} \times \lim_{x\to 0} \frac{\sin\left[\frac{x}{2}\left(e^{-x}-e^{x}\right)\right]}{\frac{x}{2}\left(e^{-x}-e^{x}\right)} \times \lim_{x\to 0} \frac{\left(e^{x}+e^{-x}\right)\left(1-e^{2x}\right)}{\left(e^{x}\right)\quad(2x)}$$

$$= 1\times 1\times \frac{(1+1)}{1}\times(-1) = -2$$

Q.9 $\lim\limits_{x\to 0} \dfrac{\left(1-3^{x}-4^{x}+12^{x}\right)}{\sqrt{(2\cos x+7)}-3}$

Sol. $\lim\limits_{x\to 0} \dfrac{\left(4^{x}-1\right)\left(3^{x}-1\right)}{\sqrt{2\cos x+7}-3} \times \dfrac{\sqrt{2\cos x+7}+3}{\sqrt{2\cos x+7}+3}$

$$\lim_{x\to 0} \frac{\left(4^{x}-1\right)}{x}\frac{\left(3^{x}-1\right)}{x} \times \frac{x^{2}\times\left(\sqrt{2\cos x+7}+3\right)}{(2\cos x+7-9)}$$

$$\lim_{x\to 0} \frac{\left(4^{x}-1\right)}{x}\frac{\left(3^{x}-1\right)}{x} \times \frac{x^{2}\times\left(\sqrt{2\cos x+7}+3\right)}{-2(1-\cos x)}$$

$$\lim_{x\to 0} \frac{\left(4^{x}-1\right)}{x}\frac{\left(3^{x}-1\right)}{x} \times \frac{\left(\frac{x^{2}}{4}\right)}{(-)\sin^{2}\frac{x}{2}} \times \left(\sqrt{2\cos x+7}+3\right)$$

$$= (\ln 4)(\ln 3)\times(-1)\times\left(\sqrt{9}+3\right)$$

$$= (-)2(\ln 2)(\ln 3)6$$

$$= -12(\ln 2)(\ln 3)$$

Q.10 If $\lim\limits_{x\to 1} \dfrac{\frac{\pi}{4}-\tan^{-1}x}{e^{\sin(\ln x)}-x^{n}}$ exists and has the value equal to $\dfrac{1}{8}$, then find n.

Sol. Apply L' Hospital rule

$$\lim_{x\to 1} \frac{-\frac{1}{1+x^{2}}}{\frac{e^{\sin(\ln x)}\cos(\ln x)}{x}-nx^{n-1}} = \frac{1}{8}$$

$$\Rightarrow \frac{-\frac{1}{2}}{1-n} = \frac{1}{8} \Rightarrow \frac{1}{n-1} = \frac{1}{4} \Rightarrow n = 5$$

LIMITS OF THE FORM $\lim_{x\to a}(f(x))^{g(x)}$

Form : $0^0, \infty^0$

Let $L = \lim_{x\to a}(f(x))^{g(x)}$

$$\Rightarrow \log_e L = \log_e\left[\lim_{x\to a}(f(x))^{g(x)}\right]$$

$$= \lim_{x\to a} g(x)\log_e\left[f(x)\right]$$

Form : 1^∞

1. $$\lim_{x\to 0}(1+x)^{\frac{1}{x}} = e \quad \text{or} \quad \lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e$$

Proof: Let $y = \lim_{x\to 0}(1+x)^{\frac{1}{x}}$

take ln on both side

$$\ln y = \lim_{x\to 0}\ln(1+x)^{\frac{1}{x}}$$

$$\ln y = \lim_{x\to 0}\frac{\ln(1+x)}{x}$$

$$\ln y = 1$$

$$y = e$$

2. $L = \lim_{x\to a} f(x)^{g(x)}$ if $\lim_{x\to a} f(x) = 1$ and $\lim_{x\to a} g(x) = \infty$

Then $L = \lim_{x\to a} f(x)^{g(x)}$

$$= \lim_{x\to a}\left(1+\left(f(x)-1\right)\right)^{\frac{1}{f(x)-1}(f(x)-1)\times g(x)}$$

$$= \left[\lim_{x\to a}\left(\left(1+\left(f(x)-1\right)\right)^{\frac{1}{f(x)-1}}\right)\right]^{\lim_{x\to a}(f(x)-1)\times g(x)} = e^{\lim_{x\to a}(f(x)-1)\times g(x)}$$

Q.1 **Evaluate** $\lim_{x\to\infty}\left(\frac{x+6}{x+1}\right)^{x+4}$.

Sol. $e^{\lim_{x\to\infty}(x+4)\left[\frac{x+6-(x+1)}{(x+1)}\right]}$

$= e^{\lim_{x\to\infty}\frac{(x+4)\times 5}{(x+1)}} = e^{\lim_{x\to\infty}\frac{5x}{x}\left[\frac{1+\frac{4}{x}}{1+\frac{1}{x}}\right]} = e^{5\times 1} = e^5$

Q.2 **Evaluate** $\lim_{x\to\infty}\left(\frac{x+2}{x+1}\right)^{x+3}$.

Sol. $e^{\lim_{x\to\infty}(x+3)\left[\frac{x+2}{x+1}-1\right]} = e^{\lim_{x\to\infty}\left[\frac{x+3}{x+1}\right]}$

$= e^{\lim_{x\to\infty}\frac{x\left(1+\frac{3}{x}\right)}{x\left(1+\frac{1}{x}\right)}}$

$= e^1 = e$

Q.3 **Evaluate** $\lim_{x\to 0}(1+x)^{\operatorname{cosec} x}$.

Sol. $\lim_{x\to 0}(1+x)^{\operatorname{cosec} x} = \lim_{x\to 0}\left[(1+x)^{\frac{1}{x}}\right]^{\frac{x}{\sin x}} = \left[\lim_{x\to 0}(1+x)^{\frac{1}{x}}\right]^{\lim_{x\to 0}\frac{x}{\sin x}} = e^1 = e$

Q.4 **Evaluate** $\lim_{x\to 0}(\cos x)^{\cot x}$.

Sol. $\lim_{x\to 0}(\cos x)^{\cot x}$

$= \lim_{x\to 0}\left[(1+(\cos x-1))^{\frac{1}{\cos x-1}}\right]^{\frac{\cos x-1}{\tan x}}$

$= \left[\lim_{x\to 0}(1+(\cos x-1))^{\frac{1}{\cos x-1}}\right]^{\lim_{x\to 0}\frac{\cos x-1}{\tan x}} = e^{\lim_{x\to 0}\frac{\cos x-1}{\sin x}\cos x}$

$= e^{\lim_{x\to 0}\frac{\cos x-1}{\sin^2 x}\cos x\sin x} = e^{\lim_{x\to 0}\left(\frac{\cos x-1}{1-\cos^2 x}.\cos x.\sin x\right)}$

$= e^{-\lim_{x\to 0}\frac{\sin x\cos x}{1+\cos x}} = e^0 = 1$

Q.5 **Evaluate** $\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\left(\frac{\sin x}{x-\sin x}\right)}$

Sol. Since $\lim_{x\to 0}\frac{\sin x}{x}=1$ and $\lim_{x\to 0}\frac{\sin x}{x-\sin x}=\lim_{x\to 0}\frac{1}{\left(\frac{x}{\sin x}-1\right)}=\frac{1}{1-1}=\infty$

$$\Rightarrow \lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\left(\frac{\sin x}{x-\sin x}\right)}=e^{\lim_{x\to 0}\left(\frac{\sin x}{x}-1\right)\left(\frac{\sin x}{x-\sin x}\right)}=e^{-\lim_{x\to 0}\frac{\sin x}{x}}=e^{-1}=\frac{1}{e}$$

Q.6 **Evaluate** $\lim_{x\to 0}\left(\frac{a^x+b^x+c^x}{3}\right)^{\frac{2}{x}};(a,b,c>0).$

Sol. We have $\lim_{x\to 0}\left(\frac{a^x+b^x+c^x}{3}\right)^{\frac{2}{x}}$

$$=e^{\lim_{x\to 0}\left(\frac{a^x+b^x+c^x}{3}-1\right)\frac{2}{x}}=e^{\frac{2}{3}\lim_{x\to 0}\left(\frac{a^x+b^x+c^x-3}{x}\right)}$$

$$=e^{\frac{2}{3}\lim_{x\to 0}\left(\frac{a^x-1}{x}+\frac{b^x-1}{x}+\frac{c^x-1}{x}\right)}=e^{\frac{2}{3}\left\{\lim_{x\to 0}\frac{a^x-1}{x}+\lim_{x\to 0}\frac{b^x-1}{x}+\lim_{x\to 0}\frac{c^x-1}{x}\right\}}$$

$$=e^{\left(\frac{2}{3}\right)[\ln a+\ln b+\ln c]}=e^{\left(\frac{2}{3}\right)\ln(abc)}=e^{\ln(abc)^{\frac{2}{3}}}=(abc)^{\frac{2}{3}}$$

Q.7 **Evaluate** $\lim_{x\to\infty}\left(\frac{1^{\frac{1}{x}}+2^{\frac{1}{x}}+3^{\frac{1}{x}}+\ldots+n^{\frac{1}{x}}}{n}\right)^{nx}$, $n\in N.$

Sol. Put $x=\frac{1}{y}$

$$\lim_{x\to\infty}\left(\frac{1^{\frac{1}{x}}+2^{\frac{1}{x}}+3^{\frac{1}{x}}+\ldots+n^{\frac{1}{x}}}{n}\right)^{nx}$$

$$=\lim_{y\to 0}\left(\frac{1^y+2^y+\ldots+n^y}{n}\right)^{\frac{n}{y}}$$

$$=e^{\lim_{y\to 0}\frac{n}{y}\left(\frac{1^y+2^y+\ldots+n^y-n}{n}\right)}$$

$$=e^{\lim_{y\to 0}\left(\frac{(1^y-1)}{y}+\frac{(2^y-1)}{y}+\ldots+\frac{(n^y-1)}{y}\right)}$$

$$=e^{[\ln 1+\ln 2+\ldots+\ln n]}$$

$$=e^{\log_e n!}=n!$$

Q.8 **Evaluate** $\lim_{x\to\infty}\left(1+\frac{1}{a+bx}\right)^{c+dx}$ **where a, b, c and d are positive.**

Sol. $\lim_{x\to\infty}\left(1+\frac{1}{a+bx}\right)^{c+dx}$

$= e^{\lim_{x\to\infty}(c+dx)\left[1+\frac{1}{a+bx}-1\right]}$

$= e^{\lim_{x\to\infty}(c+dx)\left[\frac{1}{a+bx}\right]} = e^{\lim_{x\to\infty}\frac{x\left[d+\frac{c}{x}\right]}{x\left[b+\frac{a}{x}\right]}}$

$= e^{\frac{d}{b}}$

Q.9 **Evaluate** $\lim_{x\to0}\left(\sin^2\frac{\pi}{2-ax}\right)^{\sec^2\frac{\pi}{2-bx}}$.

Sol. $\lim_{x\to0}\left(\sin^2\frac{\pi}{2-ax}\right)^{\sec^2\frac{\pi}{2-bx}}$

$= e^{\left(\lim_{x\to0}\sec^2\left(\frac{\pi}{2-bx}\right)\left[\sin^2\left(\frac{\pi}{2-ax}\right)-1\right]\right)}$

$= e^{\lim_{x\to0}(-)\left[\frac{\cos\left(\frac{\pi}{2-ax}\right)}{\cos\left(\frac{\pi}{2-bx}\right)}\right]^2}$

$= e^{(-)\lim_{x\to0}\left[\frac{-\sin\left(\frac{\pi}{2-ax}\right)\times\pi(-)(2-ax)^{-2}\times(-a)}{-\sin\left(\frac{\pi}{2-bx}\right)\pi(-)(2-bx)^{-2}\times(-b)}\right]^2}$

$= e^{-\frac{a^2}{b^2}}$

L' HOSPITAL'S RULE FOR EVALUATING LIMITS :

Rule :

If $\lim_{x\to a}\frac{f(x)}{g(x)}$ **takes** $\frac{0}{0}$ **or** $\frac{\infty}{\infty}$ **form, then,**

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}$$

Q.1 **Evaluate** $\lim_{x\to 0} \log_{\tan^2 x}\left(\tan^2 2x\right)$.

Sol. $L = \lim_{x\to 0} \dfrac{\log\left(\tan^2 2x\right)}{\log\left(\tan^2 x\right)} \left(\dfrac{\infty}{\infty}\text{ form}\right)$

Using L' hospital Rule

We have $L = \lim_{x\to 0} \dfrac{\left(\dfrac{1}{\tan^2 2x} 2\tan 2x \sec^2 2x\right)\times 2}{\dfrac{1}{\tan^2 x} 2\tan x \sec^2 x}$

$$= \lim_{x\to 0} \frac{2\left(\dfrac{1}{\sin 2x\cos 2x}\right)}{\dfrac{1}{\sin x\cos x}} = \lim_{x\to 0} \frac{\left(\dfrac{1}{\sin 2x\cos 2x}\right)}{\left(\dfrac{1}{\sin 2x}\right)} = 1\lim_{x\to 0} \frac{1}{\cos 2x} = 1$$

Q.2 **Evaluate** $\lim_{x\to 0^+} x^m (\log x)^n$, $m, n \in N$.

Sol. $\lim_{x\to 0^+} x^m (\log x)^n = \lim_{x\to 0^+} \dfrac{(\log x)^n}{x^{-m}} \left(\dfrac{\infty}{\infty}\text{ form}\right)$

$$= \lim_{x\to 0^+} \frac{n(\log x)^{(n-1)} \frac{1}{x}}{-mx^{-m-1}} \quad \text{(using L' hospital Rule)}$$

$$= \lim_{x\to 0^+} \frac{n(\log x)^{(n-1)}}{-mx^{-m}} \left(\frac{\infty}{\infty}\text{ form}\right)$$

$$= \lim_{x\to 0^+} \frac{n(n-1)(\log x)^{(n-2)} \frac{1}{x}}{\left(-m^2\right)x^{-m-1}} \quad \text{(using L'hospital Rule)}$$

$$= \lim_{x\to 0^+} \frac{n(n-1)(\log x)^{(n-2)}}{m^2 x^{-m}} \left(\frac{\infty}{\infty}\text{ form}\right)$$

$$= \lim_{x\to 0^+} \frac{n!}{(-m)^n x^{-m}} = 0 \quad \text{(differentiating N' and D' n times)}$$

Q.3 **Evaluate** $\lim_{x\to 0} \dfrac{\sin^{-1}x - \tan^{-1}x}{x^3}$.

Sol. $\lim_{x\to 0} \dfrac{\sin^{-1}x - \tan^{-1}x}{x^3}$

$$= \lim_{x\to 0} \frac{\left(1+x^2\right)-\sqrt{1-x^2}}{3x^2\sqrt{1-x^2}\left(1+x^2\right)}$$ (Using L' hospital's Rule)

$$= \lim_{x\to 0} \frac{\left(1+x^2\right)^2-\left(1-x^2\right)}{3x^2\sqrt{1-x^2}\left(1+x^2\right)} \times \frac{1}{\left(1+x^2\right)+\sqrt{1-x^2}}$$ (Rationalizing)

$$= \lim_{x\to 0} \frac{x^4+3x^2}{3x^2\sqrt{1-x^2}\left(1+x^2\right)} \times \frac{1}{\left(1+x^2\right)+\sqrt{1-x^2}}$$

$$= \lim_{x\to 0} \frac{x^2+3}{3\sqrt{1-x^2}\left(1+x^2\right)} \times \frac{1}{\left(1+x^2\right)+\sqrt{1-x^2}} = \frac{1}{2}$$

LIMITS OF FUNCTIONS HAVING BUILT IN LIMIT WITH THEM :

Examples :

$$\lim_{n\to\infty} a^n = \begin{cases} 0, & 0<a<1 \\ 1, & a=1 \\ \infty, & a>1 \end{cases}, \quad \lim_{n\to-\infty} a^n = \begin{cases} \infty, & 0<a<1 \\ 1, & a=1 \\ 0, & a>1 \end{cases}$$

Q.1 $f(x) = \lim_{n\to\infty} \frac{\tan \pi x^2 + (x+1)^n \sin x}{x^2+(x+1)^n}$, **find** $\lim_{x\to 0} f(x)$.

Sol. $f(x) = \begin{cases} \sin x, & x > 0 \\ \frac{\tan \pi x^2}{x^2}, & -1 < x < 0 \end{cases}$

$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \sin x = 0$

$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{\tan \pi x^2}{x^2} = \pi$

$\Rightarrow$ LHL $\neq$ RHL $\Rightarrow \lim_{x\to 0} f(x) =$ DNE.

Q.2 $f(x) = \lim_{n\to\infty} \frac{\cos \pi x - x^{2n} \sin(x-1)}{1+x^{2n+1}-x^{2n}}$, **find** $\lim_{x\to 1^+} f(x)$.

Sol. $f(x) = \lim_{n\to\infty} \frac{\cos \pi x - x^{2n} \sin(x-1)}{1+x\cdot x^{2n}-x^{2n}}$

when $0 < x^2 < 1$, $\lim_{n\to\infty} x^{2n} \to 0$

$\therefore$ $f(x) = \cos\pi x$

when $x^2 = 1$, $\lim\limits_{n\to\infty} x^2 \to 1$

$$\therefore\ f(x) = \frac{\cos\pi x - \sin(x-1)}{1 + x - 1} = \frac{\cos\pi x - \sin(x-1)}{x}$$

when $x^2 > 1$

$$\lim_{n\to\infty} x^{2n} \to \infty$$

$$\therefore\ \lim_{n\to\infty} \frac{1}{x^{2n}} \to 0$$

$$f(x) = \lim_{n\to\infty} \frac{\left(\frac{\cos\pi x}{x^{2n}} - \sin(x-1)\right)}{\frac{1}{x^{2n}} + x - 1} = \frac{0 - \sin(x-1)}{0 + x - 1} = \frac{-\sin(x-1)}{x-1}$$

$$f(x) = \begin{cases} \cos\pi x, & 0 < x^2 < 1 \\ \dfrac{\cos\pi x - \sin(x-1)}{x}, & x^2 = 1 \\ \dfrac{-\sin(x-1)}{x-1}, & x^2 > 1 \end{cases}$$

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} \cos\pi x = -1, \qquad \lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} \frac{-\sin(x-1)}{(x-1)} = -1$$

$\therefore \lim\limits_{x\to 1} f(x) = 1$.

Q.3 **Let $f(x) = \min\left(1,\ x^{2n},\ x^{2n+1}\right), n \in N$. The value of $\lim\limits_{x\to 0} \dfrac{e^{\tan(f(x))} - e^{\sin(f(x))}}{\tan\left(f(x)\right) - \sin\left(f(x)\right)}$, is equal to**

(A) 0 **(B) 1** **(C) 2** **(D) does not exist.**

Sol. **(B)**

$$f(x) = \min.\left(1, x^{2n}, x^{2n+1}\right), n \in N$$

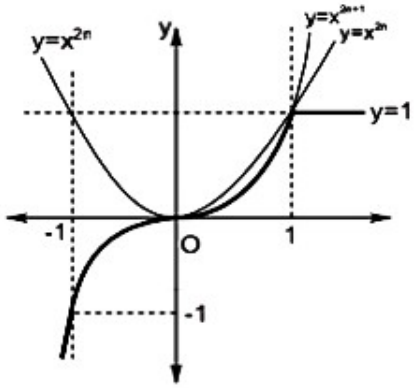
$$= \begin{cases} x^{2n+1}, & x \le 1 \\ 1, & x > 1 \end{cases}$$

$$\lim_{x\to 0}\frac{e^{\tan(f(x))}-e^{\sin(f(x))}}{\tan(f(x))-\sin(f(x))}$$

$$=\lim_{x\to 0}\frac{e^{\sin(f(x))}\left(e^{\tan(f(x))-\sin(f(x))}-1\right)}{\tan(f(x))-\sin(f(x))}$$

$$=\lim_{x\to 0}\frac{\left(e^{\tan(x^{2n+1})-\sin(x^{2n+1})}-1\right)}{\tan(x^{2n+1})-\sin(x^{2n+1})}=\lim_{y\to 0}\frac{e^{y}-1}{y}=1$$

where $y=\tan(x^{2n+1})-\sin(x^{2n+1})$



Q.4 **Let** $f(x)=\lim_{n\to\infty}\frac{\sin(\pi x^4)+(x+2)^n.\frac{\tan\pi x}{x+1}}{1+(x+2)^n-x^4}$ **, then** $\lim_{x\to -1}f(x)$ **is equal to**

(A) π **(B)** $\frac{22}{7}$ **(C) 1** **(D) Non existent**

Sol. **(A)**

$$\text{Let } f(x)=\lim_{n\to\infty}f(x)\frac{\sin(\pi x^4)+(x+2)^n\frac{\tan\pi x}{(x+1)}}{1+(x+2)^n-x^4}$$

$$f(x)=\begin{cases}\dfrac{\sin(\pi x^4)}{1-x^4}, & x<-1\\ \dfrac{\tan\pi x}{x+1}, & x>-1\end{cases}$$

$$\lim_{x\to -1}f(x)=\lim_{x\to -1^-}\frac{\tan\pi x}{(x+1)}=\lim_{x\to -1^+}\frac{(\tan\pi(1+x))}{(1+x)}=\pi$$

$$\lim_{x\to -1^-}f(x)=\lim_{x\to -1^-}\frac{\sin(\pi x^4)}{(1-x^4)}=\lim_{x\to -1^-}\frac{\sin(\pi(1-x^4))}{(1-x^4)}=\pi$$

$$\lim_{x\to -1^-}f(x)=\pi$$

EXPANSION OF FUNCTION :

Expansion of function like Binomial expansion, exponential and logarithmic expansion, expansion of sinx, cosx, tanx should be remembered & are given below:

(i) $a^x = 1 + \frac{x \ln a}{1!} + \frac{x^2 \ln^2 a}{2!} + \frac{x^3 \ln^3 a}{3!} + \ldots \ (a > 0)$

(ii) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots \forall x \in R$

(iii) $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots \ \text{for} - 1 < x \leq 1$

(iv) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots \ \left(-\frac{\pi}{2} < x < \frac{\pi}{2}\right)$

(v) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots \ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(vi) $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \ldots \ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(vii) $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots$

(viii) $\sin^{-1} x = x + \frac{1^2}{3!}x^3 + \frac{1^2.3^2}{5!}x^5 + \frac{1^2.3^2.5^2}{7!}x^7 + \ldots$

(ix) $\sec^{-1} x = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \frac{61x^6}{6!} + \ldots$

Q.1 Evaluate the following limit:

(i) $\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$

Sol. $\lim_{x \to 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \to 0} \frac{\left(1 + x + \frac{x^2}{2!} + \ldots\right) - 1 - x}{x^2} = \frac{1}{2!} = \frac{1}{2}$

Q. **(ii)** $\lim_{x \to 0} \frac{x - \sin x}{x^3}$

Sol. $\lim_{x \to 0} \frac{x - \sin x}{x^3} = \lim_{x \to 0} \frac{x - \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots\right)}{x^3} = \frac{1}{3!} = \frac{1}{6}$

Q. **(iii)** $\lim\limits_{x\to0}\dfrac{e^x-e^{-x}-2x}{x^3}$

Sol. $\lim\limits_{x\to0}\dfrac{e^x-e^{-x}-2x}{x^3}=\lim\limits_{x\to0}\dfrac{\left(1+x+\dfrac{x^2}{2!}+\ldots\right)-\left(1-x+\dfrac{x^2}{2!}\ldots\right)-2x}{x^3}$

$$=\lim_{x\to0}\frac{\left(2x+\dfrac{2x^3}{3!}+\ldots\right)-2x}{x^3}=\frac{2}{3!}=\frac{1}{3}$$

Q. **(iv)** $\lim\limits_{x\to0}\dfrac{x-\tan x}{x^3}$

Sol. $\lim\limits_{x\to0}\dfrac{x-\tan x}{x^3}=\lim\limits_{x\to0}\dfrac{x-\left(x+\dfrac{x^3}{3}+\dfrac{2x^5}{15}+\ldots\right)}{x^3}=\dfrac{-1}{3}$

Q.2 **Evaluate** $\lim\limits_{x\to0}\dfrac{\tan^2x-x^2}{x^2\tan^2x}$.

Sol. $\lim\limits_{x\to0}\dfrac{\tan^2x-x^2}{x^2\tan^2x}=\lim\limits_{x\to0}\dfrac{(\tan x-x)(\tan x+x)}{x^2\tan^2x}$

$$=\lim_{x\to0}\frac{\left(x+\frac{1}{3}x^3+\ldots-x\right)(\tan x+x)}{x^4\left(\frac{\tan x}{x}\right)^2}=\frac{1}{3}\frac{\left(\frac{\tan x}{x}+1\right)}{\left(\frac{\tan x}{x}\right)^2}=\frac{1}{3}\left(\frac{1+1}{1}\right)=\frac{2}{3}$$

Q.3 **Evaluate** $\lim\limits_{x\to0}\dfrac{e^{x^3}-1-x^3}{\sin^62x}$.

Sol. $\lim\limits_{x\to0}\dfrac{e^{x^3}-1-x^3}{\sin^62x}=\lim\limits_{x\to0}\dfrac{\left(1+x^3+\dfrac{x^6}{2!}+\ldots\right)-1-x^3}{\dfrac{\sin^62x}{(2x)^6}\cdot(2x)^6}$

$$\lim_{x\to0}\frac{\frac{1}{2!}+\frac{x^3}{3!}+\ldots}{2^6}=\frac{\frac{1}{2}}{2^6}=\frac{1}{128}$$

Q.4 **Evaluate** $\lim_{x\to 0}\frac{1}{(\sin^{-1}x)^2}-\frac{1}{x^2}$.

Sol. Put $x=\sin\theta \Rightarrow \lim_{\theta\to 0}\frac{1}{\theta^2}-\frac{1}{\sin^2\theta}=\lim_{\theta\to 0}\frac{\sin^2\theta-\theta^2}{\theta^2\sin^2\theta}$

$$\lim_{\theta\to 0}\frac{(\sin\theta-\theta)(\sin\theta+\theta)}{\theta^4}=2.\lim_{\theta\to 0}\frac{\sin\theta-\theta}{\theta^3}=-\frac{1}{3}$$

Don't do it :

$$\lim_{x\to 0}\frac{1}{(\sin^{-1}x)^2}-\frac{1}{x^2}=\lim_{x\to 0}\frac{1}{x^2}.\frac{x^2}{(\sin^{-1}x)^2}-\frac{1}{x^2}=\lim_{x\to 0}\frac{1}{x^2}-\frac{1}{x^2}=0\text{ , is wrong.}$$

Q.5 **Evaluate** $\lim_{x\to\infty} x-x^2\ell n\left(1+\frac{1}{x}\right)$.

Sol. $\lim_{x\to\infty} x-x^2\ell n\left(1+\frac{1}{x}\right)$ put $x=\frac{1}{y}$

$$=\lim_{y\to 0}\frac{1}{y}-\frac{\ln(1+y)}{y^2}=\lim_{y\to 0}\frac{y-\ln(1+y)}{y^2}=\lim_{y\to 0}\frac{y-\left(y-\frac{y^2}{2}+\frac{y^3}{3!}-\ldots\right)}{y^2}=\frac{1}{2}$$

Don't do it :

$$\lim_{y\to 0}\left(\frac{1}{y}-\frac{\ln(1+y)}{y}.\frac{1}{y}\right)=\frac{1}{y}-\frac{1}{y}=0\text{ as }\lim_{y\to 0}\frac{\ln(1+y)}{y}=1\text{, is not correct.}$$

Q.6 **Evaluate the following limits :**

(i) $\lim_{x\to 0}\frac{e^{x^2}-\cos x}{x^2}$

Sol. $\lim_{x\to 0}\frac{e^{x^2}-\cos x}{x^2}=\lim_{x\to 0}\frac{1+x^2+\frac{(x^2)^2}{2!}+\frac{(X^2)^3}{3!}-\left(1-\frac{x^2}{2!}+\ldots\right)}{x^2}=\frac{3}{2}$

Q. **(ii)** $\lim_{x\to 0}\frac{e^{\sin 2x}-e^{\sin x}}{x}$

Sol. $\lim_{x\to 0}\frac{e^{\sin 2x}-e^{\sin x}}{x}=\lim_{x\to 0}\frac{\left(1+\sin 2x+\frac{(\sin 2x)^2}{2!}+\ldots\right)-(1+\sin x+\ldots)}{x}$

$$=\lim_{x\to 0}\frac{\sin 2x-\sin x}{x}=2-1=1$$

Q.7 **Evaluate** $\lim\limits_{x\to0}\dfrac{(1+x)^{\frac{1}{x}}-e}{x}$

Sol. $\lim\limits_{x\to0}\dfrac{(1+x)^{\frac{1}{x}}-e}{x}=\lim\limits_{x\to0}\dfrac{e^{\frac{1}{x}\ln(1+x)}-e}{x}=\lim\limits_{x\to0}\dfrac{e\left(e^{\frac{\ln(1+x)-x}{x}}-1\right)}{\left(\frac{\ln(1+x)-x}{x}\right)}\cdot\left(\dfrac{\ln(1+x)-x}{x^2}\right)$

$$=e\lim_{x\to0}\frac{\left(e^{\frac{\ln(1+x)}{x}-1}-1\right)}{\left(\frac{\ln(1+x)}{x}-1\right)}\lim_{x\to0}\left(\frac{\ln(1+x)-x}{x^2}\right)=e(1).\left(\frac{-1}{2}\right)=\frac{-e}{2}$$

Don't do it :

$$\lim_{x\to0}\frac{(1+x)^{1/x}-e}{x}=l$$

Let $x=\dfrac{y}{2}$

$$l=\lim_{y\to0}\frac{\left(1+\frac{y}{2}\right)^{\frac{2}{y}}-e}{\frac{y}{2}}=2\cdot\lim_{y\to0}\frac{\left(1+\frac{y^2}{4}+y\right)^{\frac{1}{y}}-e}{y}=2\cdot\lim_{y\to0}\frac{e^{\frac{1}{y}\left(1+\frac{y^2}{4}+y-1\right)}-e}{y}\ *$$

$$=2\lim_{y\to0}\frac{e^{\left(\frac{y^2}{4}+y\right)\frac{1}{y}}-e}{y}=2.\lim_{y\to0}\frac{e^{\frac{y}{4}+1}-e}{y}=2.e\lim_{y\to0}\frac{e^{\frac{y}{4}}-1}{\frac{y}{4}}.\frac{1}{4}=\frac{2e}{4}=\frac{e}{2}$$

Note that mistake occurred at *.

Q.8 **If** $\lim\limits_{x\to0}\dfrac{A\cos x+Bx\sin x-5}{x^4}$ **exists and finite. Find A & B and also the limit.**

Sol. Let $L=\lim\limits_{x\to0}\dfrac{A\cos x+Bx\sin x-5}{x^4}$

$$=\lim_{x\to0}\frac{A\left(1-\frac{x^2}{2!}+\frac{x^4}{4!}\ldots\right)+Bx\left(x-\frac{x^3}{3!}+\ldots\right)-5}{x^4}$$

$$L=\lim_{x\to0}\frac{(A-5)+\left(B-\frac{A}{2}\right)x^2+\left(\frac{A}{24}-\frac{B}{6}\right)x^4}{x^4}=\text{ finite value}$$

$\Rightarrow A=5, B=\dfrac{A}{2}=\dfrac{5}{2}, L=\dfrac{A}{24}-\dfrac{B}{6}=\dfrac{-5}{24}$.

Q.9 **Let** $f(x)=\dfrac{4+\sin 2x+A\sin x+B\cos x}{x^2}$. **If** $\lim\limits_{x\to 0} f(x)$ **exists and finite find A, B and the limit.**

Sol. Let $L=\lim\limits_{x\to 0}\dfrac{4+\sin 2x+A\sin x+B\cos x}{x^2}$

$$L=\lim_{x\to 0}\frac{4+\left(2x-\frac{(2x)^3}{3!}+\ldots\right)+A\left(x-\frac{x^3}{3!}+\ldots\right)+B\left(1-\frac{x^2}{2!}+\frac{x^4}{4!}+\ldots\right)}{x^2}$$

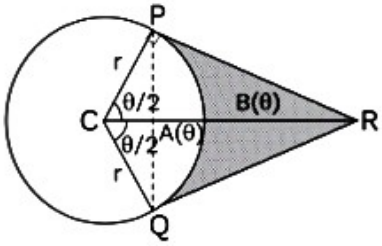
$$L=\lim_{x\to 0}\frac{(4+B)+(A+2)x-\frac{B}{2}x^2+\ldots}{x^2}$$

$\Rightarrow B=-4, A=-2, L=2.$

Q.10 An arc PQ of a circle subtends a central angle θ as shown. Let $A(\theta)$ be the area between the chord PQ and the arc PQ. Let $B(\theta)$ be the area between the tangent lines PR and QR and the arc PQ. Find $\lim\limits_{\theta\to 0}\dfrac{A(\theta)}{B(\theta)}$.

Sol. $A(\theta)$ = Area of sector PCQP - Area of ΔPCQ

$$=\frac{1}{2}r^2\theta-\frac{1}{2}r^2\sin\theta=\frac{1}{2}r^2(\theta-\sin\theta)$$



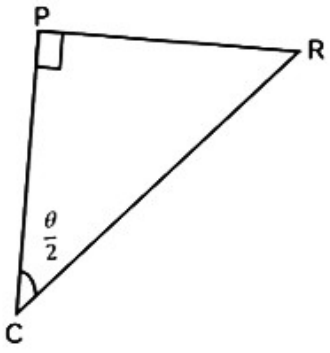
$B(\theta)$ = Area of quadrilateral PCQR − Area of sector PCQP

= 2(Area of ΔCPR) − Area of sector PCQP

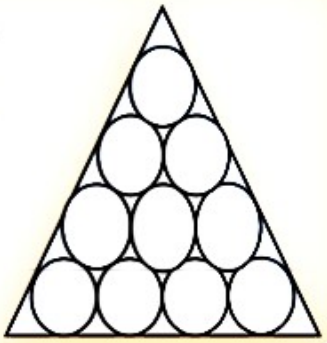
$$=2\left(\frac{1}{2}r^2.\tan\frac{\theta}{2}\right)-\frac{1}{2}r^2\theta=\frac{r^2}{2}\left(2\tan\frac{\theta}{2}-\theta\right)$$

$$\lim_{\theta\to 0}\frac{A(\theta)}{B(\theta)}=-\lim_{\theta\to 0}\frac{\theta-\sin\theta}{2\tan\left(\frac{\theta}{2}\right)-\theta}$$

$$=\lim_{\theta\to 0}\frac{\theta-\left(\theta-\frac{\theta^3}{3!}+\frac{\theta^4}{4!}\ldots\right)}{2\left(\frac{\theta}{2}+\frac{\left(\frac{\theta}{2}\right)^3}{3}+\frac{2\left(\frac{\theta}{2}\right)^5}{15}\ldots\right)-\theta}=\frac{\frac{1}{3!}}{2.\frac{1}{3.2^3}}=\frac{3.8}{2.6}=2$$



Q.11 Suppose that circle of equal diameter are packed tightly in a rows inside an equilateral triangle. (The figure illustrates the case n = 4.) If A is the area of the triangle and A_n is the total area occupied by the circles in n rows



then $\lim_{n\to\infty}\frac{A_n}{A}$ equals

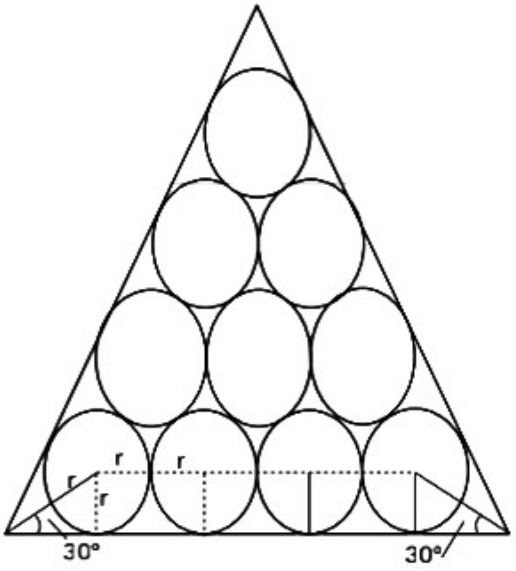
(A) $\frac{\pi}{\sqrt{3}}$ (B) $\frac{\pi\sqrt{3}}{6}$ (C) $\frac{\pi}{2\sqrt{3}}$ (D) $\frac{\pi}{6}$

Sol. Let radius of each circle = r and side of triangle = a

$$a = (n-2)2r + 2(r + r\cot 30^\circ)$$

$$= r(2n - 4 + 2 + 2\sqrt{3})$$

$$= r(2n - 2 + 2\sqrt{3})$$



$$A = \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4}r^2 \cdot 4(n - 1 + \sqrt{3})^2$$

$$A = \sqrt{3}r^2(n + \sqrt{3} - 1)^2$$

$$A_n = \pi r^2(1 + 2 + \ldots + n) = \frac{n(n+1)}{2}\pi r^2$$

$$\lim_{n\to\infty}\frac{A_n}{A} = \lim_{n\to\infty}\frac{\frac{n(n+1)}{2}\pi r^2}{\sqrt{3}r^2(n + \sqrt{3} - 1)^2}$$

$$= \lim_{n\to\infty}\frac{\pi}{2\sqrt{3}}\frac{\left(1 + \frac{1}{n}\right)}{\left(1 + \frac{\sqrt{3} - 1}{n}\right)^2} = \frac{\pi}{2\sqrt{3}}\frac{(1+0)}{(1+0)^2} = \frac{\pi}{2\sqrt{3}}$$